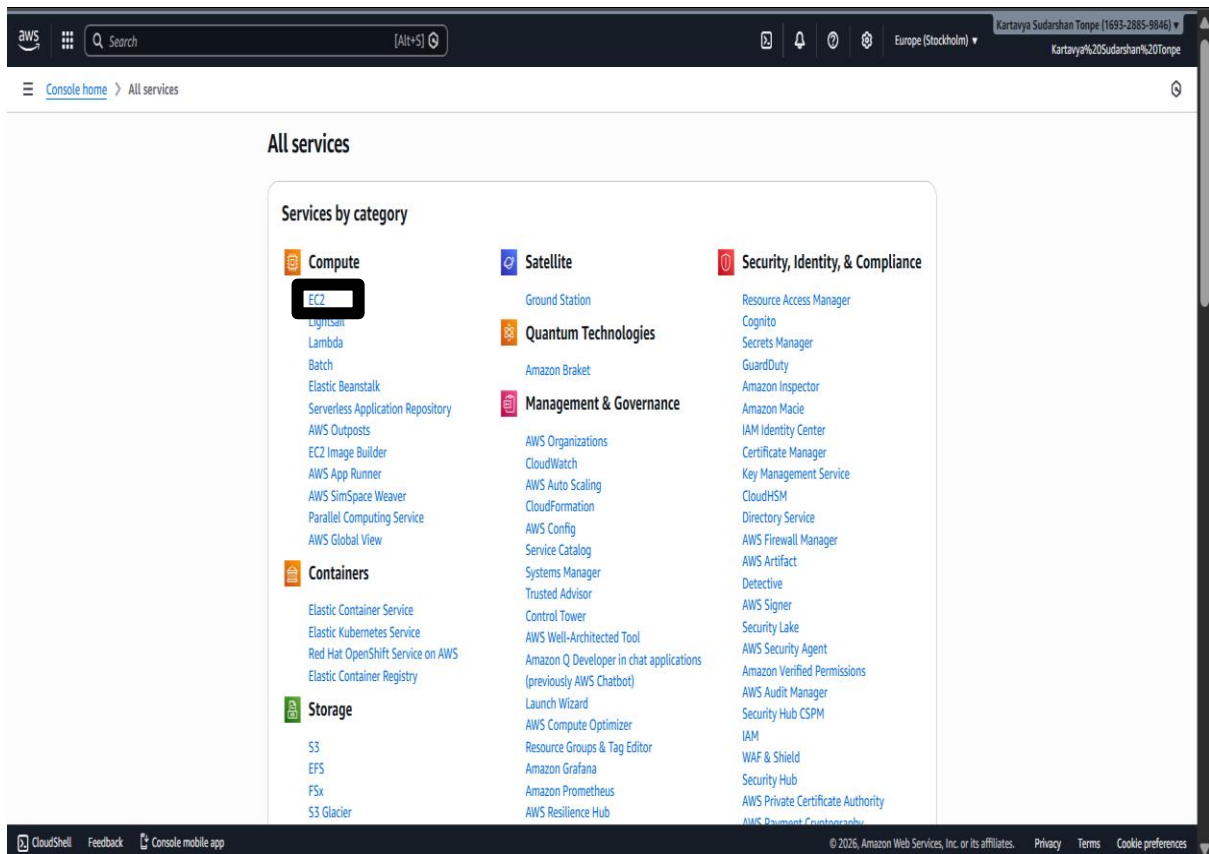


ASSIGNMENT NO:- 1

Steps to launch an Instance:-

Step 1:-Login as root user

Step 2:-Go to EC2 web service



Step 3:- Launch Instance

The screenshot shows the AWS Management Console for the EC2 service. The left-hand navigation menu includes links to Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, and Lifecycle Manager. The main content area features a large hero section titled "Amazon Elastic Compute Cloud (EC2)" with the subtitle "Create, manage, and monitor virtual servers in the cloud." Below this, there's a "Launch a virtual server" card with a prominent "Launch instance" button. Other buttons include "View dashboard", "Get started walkthroughs", and "Get started tutorial". A "Benefits and features" section highlights EC2's scalability and control, listing benefits like full control, variety of server sizes, and global availability. To the right, "Additional actions" include "View running instances" and "Migrate a server". A "Pricing (US)" section links to "EC2 pricing options". The footer shows the URL "https://eu-north-1.console.aws.amazon.com/console/home?region=eu..." and copyright information for 2026.

Step 4:- Name the Instance

The screenshot displays the "Launch an instance" wizard. At the top, a blue banner offers a walkthrough. The main heading is "Launch an instance" with an "Info" link. Below it, a sub-heading "Name and tags" is followed by a text input field containing "Kartavya1_server". To the right of the input field is an "Add additional tags" link. The "Application and OS Images (Amazon Machine Image)" section is expanded, showing a search bar and a "Quick Start" tab. Below the search bar, a grid of AMI icons is visible, including Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian. On the right side, a "Summary" panel lists the configuration: "Number of instances" set to 1, "Software Image (AMI)" as Amazon Linux 2023 AMI, "Virtual server type (instance type)" as t3.micro, "Firewall (security group)" as New security group, and "Storage (volumes)" as 1 volume(s) - 8 GiB. At the bottom right of the summary panel are "Cancel" and "Launch instance" buttons, along with a "Preview code" link. The footer includes links for CloudShell, Feedback, and Console Mobile App, along with copyright information for 2026.

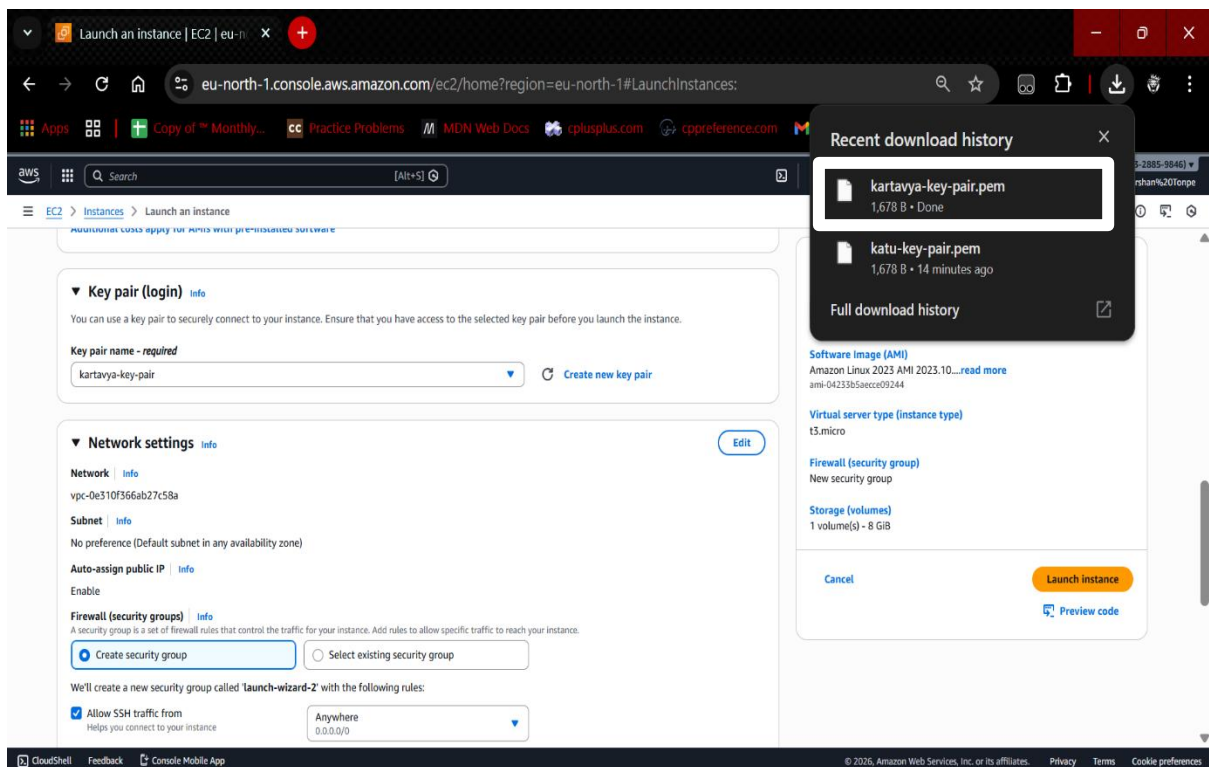
Step 5:-Choose OS

The screenshot shows the AWS Management Console interface for launching an EC2 instance. The 'Amazon Linux' AMI is selected and highlighted. The 'Summary' panel on the right provides a quick overview of the configuration: 1 instance, Amazon Linux 2023 AMI, t3.micro instance type, new security group, and 8 GiB storage. The 'Launch instance' button is prominently displayed.

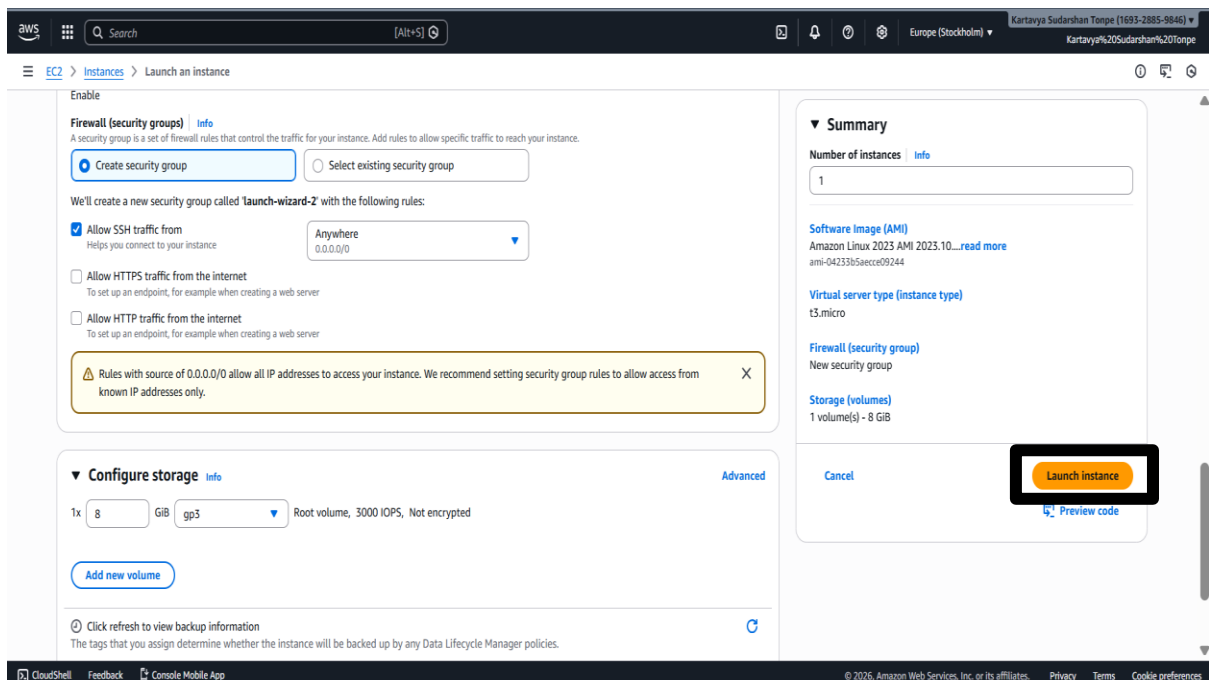
Step 6:-Create a key

The screenshot shows the 'Create key pair' dialog box overlaid on the 'Launch an instance' page. The dialog prompts for a 'Key pair name' (kartavya-key-pair), 'Key pair type' (RSA), and 'Private key file format' (.pem). A warning message states: 'When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance.' The 'Create key pair' button is highlighted.

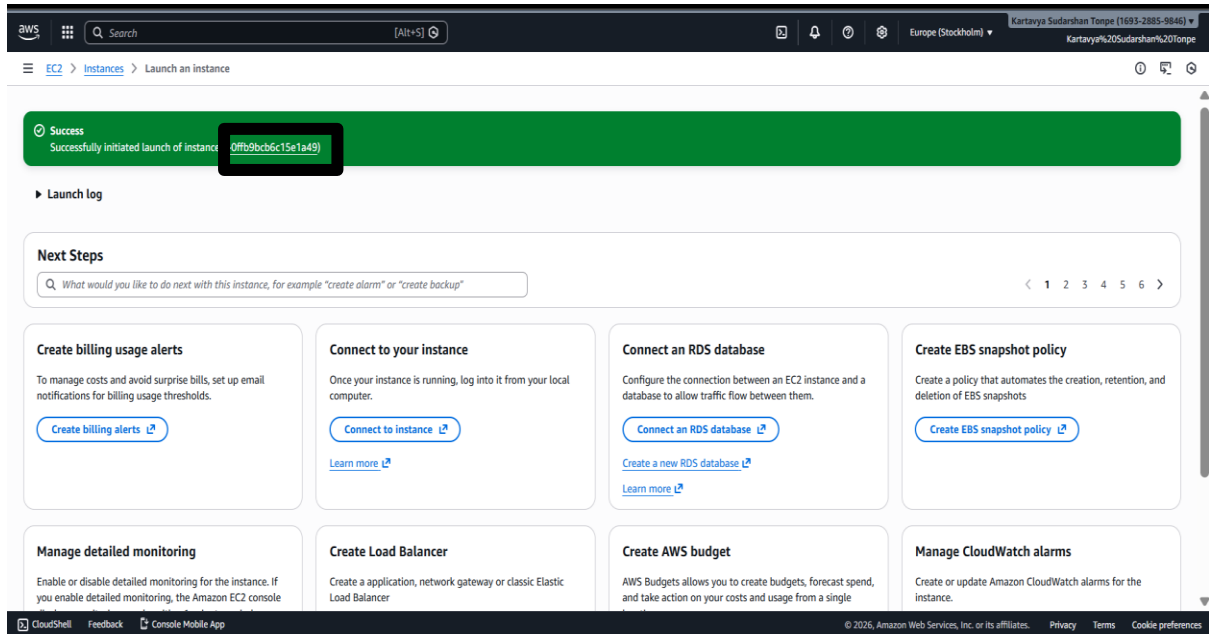
Step 7:-Save the key



Step 8:-Launch the Instance



Step 9:-Click on Instance ID



Success
Successfully initiated launch of instance **i-0ffb9bcb6c15e1a49**

Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
[Create EBS snapshot policy](#)

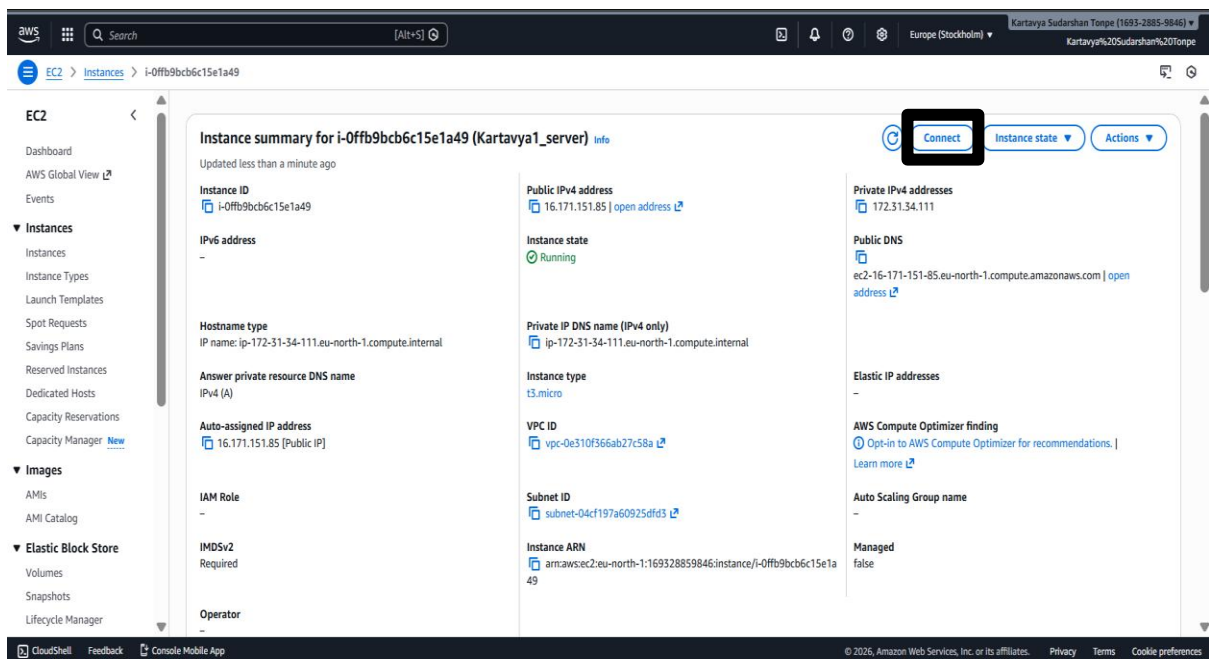
Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console

Create Load Balancer
Create a application, network gateway or classic Elastic Load Balancer

Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single

Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.

Step 10:-Connect an Instance



Instance summary for i-0ffb9bcb6c15e1a49 (Kartavya1_server)

Updated less than a minute ago

Instance ID
i-0ffb9bcb6c15e1a49

IPv6 address
-

Hostname type
IP name: ip-172-31-34-111.eu-north-1.compute.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
16.171.151.85 [Public IP]

IAM Role
-

IMDSv2
Required

Operator
-

Public IPv4 address
16.171.151.85 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-34-111.eu-north-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-0e310f366ab27c58a | [open](#)

Subnet ID
subnet-04cf197a60925df43 | [open](#)

Instance ARN
arn:aws:ec2:eu-north-1:169328859846:instance/i-0ffb9bcb6c15e1a49

Private IPv4 addresses
172.31.34.111

Public DNS
ec2-16-171-151-85.eu-north-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Step 11:-Click on Connect

Connect info

Connect to an instance using the browser-based client.

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID
i-0ffb9bcb6c15e1a49 (Kartavya1_server)

Connection type

☒ **Connect using a Public IP**
Connect using a public IPv4 or IPv6 address

☐ **Connect using a Private IP**
Connect using a private IP address and a VPC endpoint

Public IPv4 address
16.171.151.85

IPv6 address
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

[Cancel](#) [Connect](#)

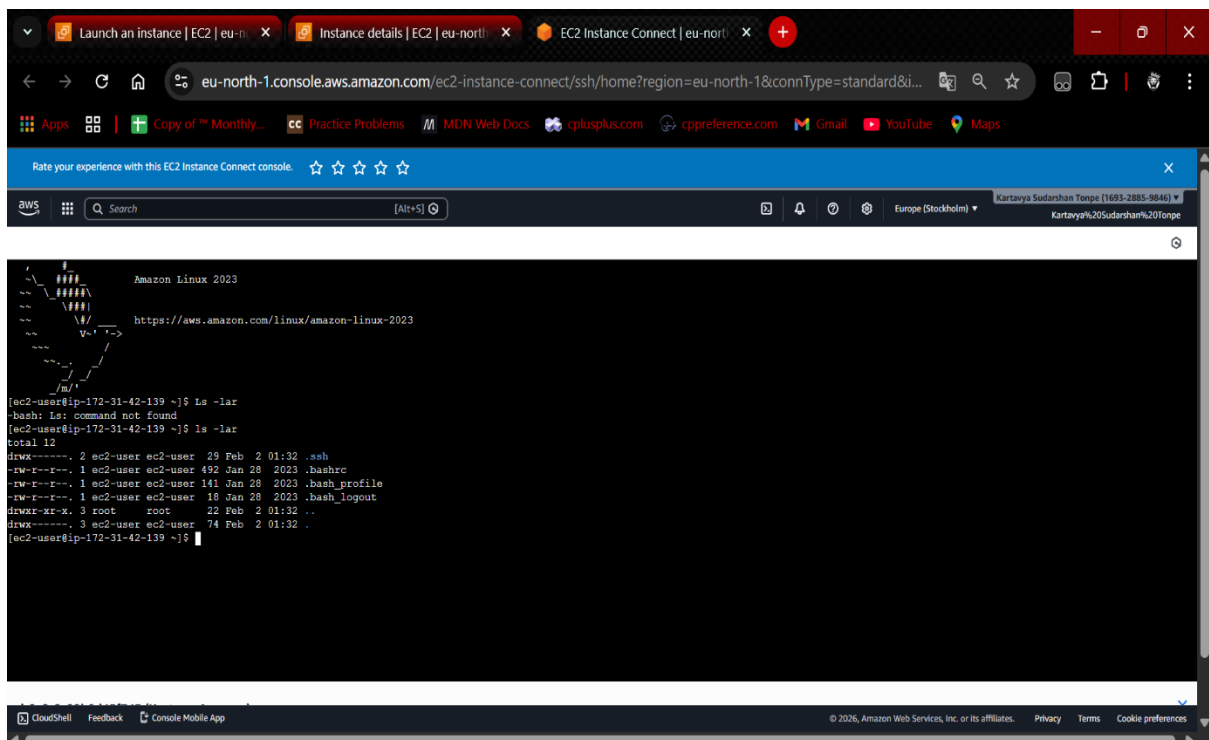
Step 12:-Type command `ls -lar`

Amazon Linux 2023

`https://aws.amazon.com/linux/amazon-linux-2023`

```
ec2-user@ip-172-31-42-139 ~]$ ls -lar
```

Step 13:-Files present By default in an Instance

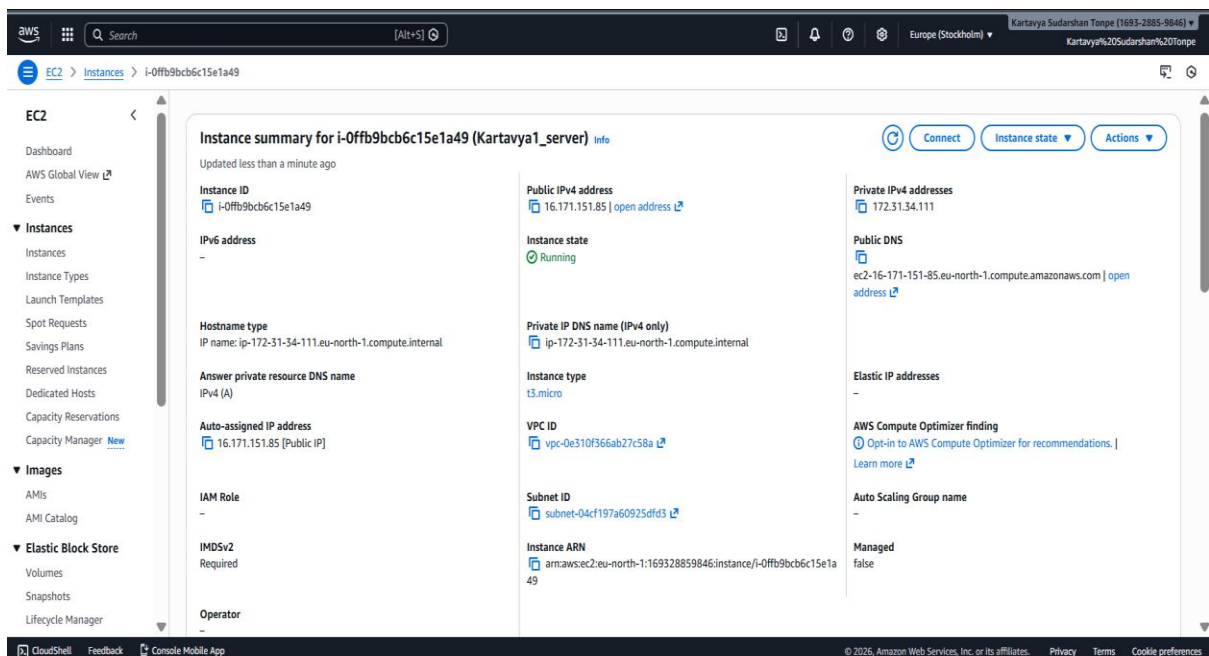


The screenshot shows the AWS EC2 Instance Connect console. The terminal window displays the following output:

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-42-139 ~]$ ls -lar
-bash: ls: command not found
[ec2-user@ip-172-31-42-139 ~]$ ls -lar
total 12
drwx----- 2 ec2-user ec2-user 29 Feb  2 01:32 .ssh
-rw-r--r-- 1 ec2-user ec2-user 490 Jan 28  2023 .bashrc
-rw-r--r-- 1 ec2-user ec2-user 141 Jan 28  2023 .bash_profile
-rw-r--r-- 1 ec2-user ec2-user 18 Jan 28  2023 .bash_logout
drwxr-xr-x 3 root    root      22 Feb  2 01:32 ..
drwx----- 3 ec2-user ec2-user 74 Feb  2 01:32 .
```

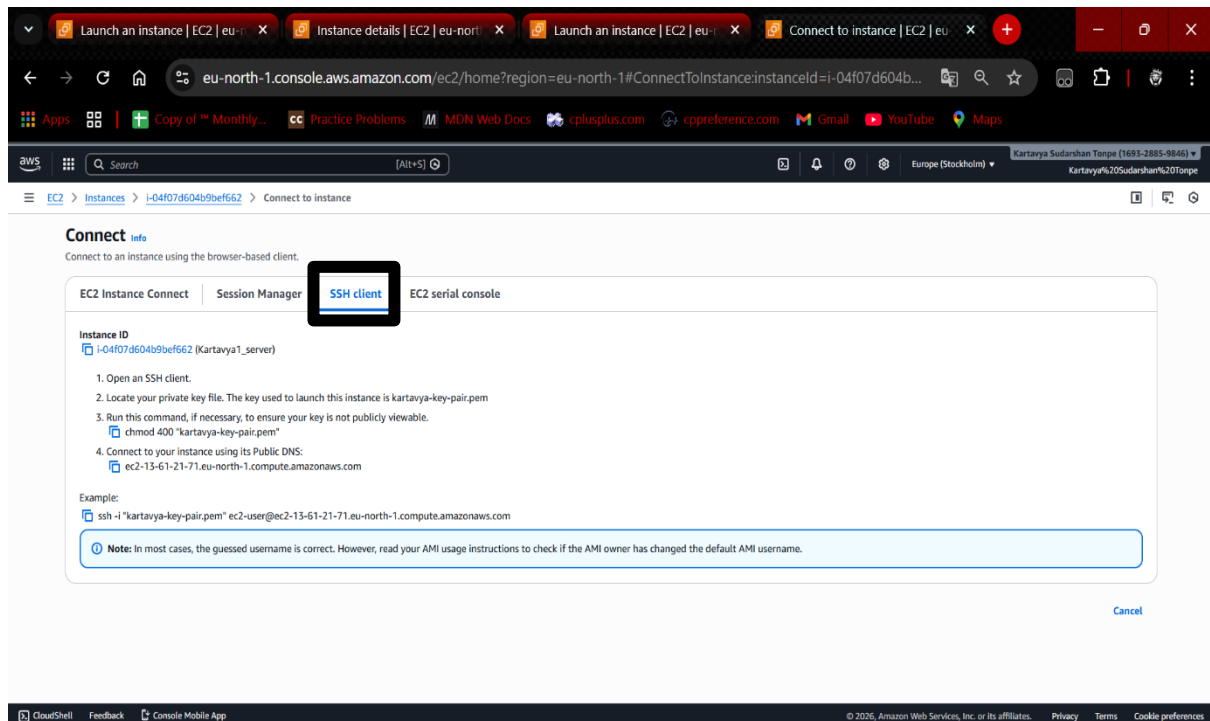
Step 14:-Connect an Instance by using another method



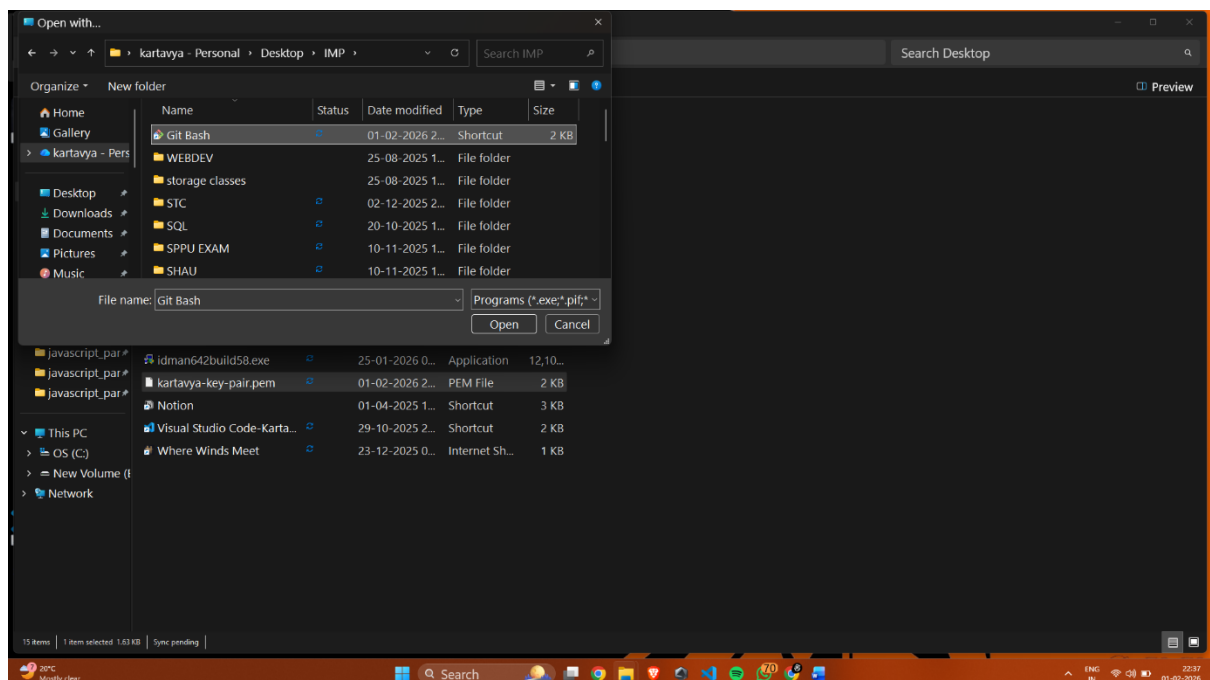
The screenshot shows the AWS Management Console for an EC2 instance. The instance is named "Kartavya1_server" and has the ID "i-0ffb9bcb6c15e1a49". The instance is in the "Running" state. The console displays various details about the instance, including its IP addresses, DNS names, and VPC configuration.

Property	Value
Instance ID	i-0ffb9bcb6c15e1a49
Public IPv4 address	16.171.151.85 open address
Private IPv4 addresses	172.31.34.111
Instance state	Running
Public DNS	ec2-16-171-151-85.eu-north-1.compute.amazonaws.com open address
Private IP DNS name (IPv4 only)	ip-172-31-34-111.eu-north-1.compute.internal
Instance type	t3.micro
VPC ID	vpc-0e310f366ab27c58a open
Subnet ID	subnet-04cf197a60925dfd3 open
Instance ARN	arn:aws:ec2:eu-north-1:169328859846:instance/i-0ffb9bcb6c15e1a49
Host name type	IP name: ip-172-31-34-111.eu-north-1.compute.internal
Answer private resource DNS name	IPV4 (A)
Auto-assigned IP address	16.171.151.85 [Public IP]
IAM Role	-
IMDSv2	Required
Operator	-
Elastic IP addresses	-
AWS Compute Optimizer finding	Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto Scaling Group name	-
Managed	false

Step 15:-Connect an Instance using SSH Client Method



Step 16:-Open kartavya-key-pair.pem file in Git Bash



Step 17:-Type commands to check the permission type of file

```
MINGW64/C:/Users/katus/downloads
katus@Kartavya MINGW64 ~/downloads
$ ls -lar
total 1886349
-rw-r--r-- 1 katus 197609 10041124 Dec 23 15:05 smart-form-filler-demo.mp4
-rw-r--r-- 1 katus 197609 10041124 Dec 23 15:05 'smart-form-filler-demo (1).mp4'
-rw-r--r-- 1 katus 197609 319472 Dec 23 15:07 smart-form-filler-1.0.0.zip
-rwxr-xr-x 1 katus 197609 28688288 Dec 13 2024 python-3.13.1-amd64.exe*
-rwxr-xr-x 1 katus 197609 26507904 Nov 23 2023 python-3.12.0-amd64.exe*
-rw-r--r-- 1 katus 197609 616451 Jan 27 2024 pxfuel.jpg
-rw-r--r-- 1 katus 197609 32239616 Oct 29 19:45 node-v24.11.0-x64.msi
-rw-r--r-- 1 katus 197609 53358592 Oct 29 19:07 mongosh-2.5.8-x64.msi
-rw-r--r-- 1 katus 197609 51255620 Oct 29 19:06 mongosh-2.5.8-win32-x64.zip
-rw-r--r-- 1 katus 197609 792559616 Oct 29 18:51 mongodb-windows-x86_64-8.2.1-signed.msi
-rw-r--r-- 1 katus 197609 16752316 Nov 23 2023 mingw-w64-x11.0.0.zip
-rwxr-xr-x 1 katus 197609 86528 Nov 23 2023 'mingw-get-setup.exe*'
-rw-r--r-- 1 katus 197609 1678 Feb 1 22:20 kartavya-key-pair.pem
-rwxr-xr-x 1 katus 197609 295320 Jul 9 2024 'dowebsetup.exe*'
-rwxr-xr-x 1 katus 197609 295320 Jul 28 2024 'dowebsetup (1).exe*'
-rwxr-xr-x 1 katus 197609 226334344 Jul 28 2024 'dotnet-sdk-8.0.303-win-x64.exe*'
-rw-r--r-- 1 katus 197609 282 Nov 10 2024 desktop.ini
-rw-r--r-- 1 katus 197609 97320 Dec 20 2024 brandonfowler.exe-runner-1.1.1.vsix
-rw-r--r-- 1 katus 197609 41705 Nov 12 12:08 be_computer-engineering_semester-5_2024_october_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 139091 Nov 12 12:08 be_computer-engineering_semester-5_2024_may_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 88143 Nov 12 12:08 be_computer-engineering_semester-5_2023_november_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 12661 Feb 20 2025 'WhatsApp Image 2025-02-20 at 17.56.00_6b9704e4.jpg'
-rwxr-xr-x 1 katus 197609 94619344 Nov 23 2023 'VSCodesetupSetup-x64-1.84.2.exe*'
-rwxr-xr-x 1 katus 197609 25397512 Jul 28 2024 'VC_redist.x64.exe*'
drwxr-xr-x 1 katus 197609 0 Dec 6 08:28 'Telegram Desktop/'
-rw-r--r-- 1 katus 197609 1175307 Nov 12 12:12 SureLock_seminar.pdf
-rw-r--r-- 1 katus 197609 140151 Jul 22 2024 Self-Declaration-x84-8mm-marathiform.pdf
drwxr-xr-x 1 katus 197609 0 Aug 4 18:57 Orientations/
-rw-r--r-- 1 katus 197609 1359228 Apr 24 2024 LBM_Updated.docx
-rwxr-xr-x 1 katus 197609 167496664 Aug 31 2024 GitHubDesktopSetup-x64.exe*
-rw-r--r-- 1 katus 197609 703657 Nov 13 13:34 G545W94ApplicationForm.pdf
-rw-r--r-- 1 katus 197609 86007808 Dec 22 18:21 EpicInstaller-19.0.0-6e1daf5a6ad8410dad4fc54e9c601d76.msi
-rwxr-xr-x 1 katus 197609 107226856 Mar 8 2025 'DiscordSetup.exe*'
-rwxr-xr-x 1 katus 197609 107226856 Mar 8 2025 'DiscordSetup (1).exe*'
drwxr-xr-x 1 katus 197609 0 Aug 4 18:52 'Day 1 WELCOME + HTML/'
-rw-r--r-- 1 katus 197609 34792458 Jul 28 2024 'DM - Technical.pdf*'
-rw-r--r-- 1 katus 197609 34792458 Aug 19 2024 'DM - Technical (1).pdf*'
-rw-r--r-- 1 katus 197609 801147 Oct 31 15:47 'DBMSL11.pdf'
-rwxr-xr-x 1 katus 197609 1382960 Nov 15 2023 'ChromeSetup.exe*'
-rw-r--r-- 1 katus 197609 7781185 Jul 23 2024 'CG ASSIGNMENT 1.pdf'
drwxr-xr-x 1 katus 197609 0 Dec 3 2023 'B9CEED6F-ASUSPCAssistant_qmba6cd70vzyy/App/'
-rw-r--r-- 1 katus 197609 7942584 Nov 18 22:40 'AnyDesk.exe*'
-rw-r--r-- 1 katus 197609 1512448 Aug 16 2024 722401.msi
drwxr-xr-x 1 katus 197609 0 Jan 2 23:09 ./
drwxr-xr-x 1 katus 197609 0 Feb 1 22:46 ./
katus@Kartavya MINGW64 ~/downloads
$
```

Step 18:- Permission of kartavya-key-pair.pem file

```
MINGW64/C:/Users/katus/downloads
katus@Kartavya MINGW64 ~/downloads
$ ls -lar
total 1886349
-rw-r--r-- 1 katus 197609 10041124 Dec 23 15:05 smart-form-filler-demo.mp4
-rw-r--r-- 1 katus 197609 10041124 Dec 23 15:05 'smart-form-filler-demo (1).mp4'
-rw-r--r-- 1 katus 197609 319472 Dec 23 15:07 smart-form-filler-1.0.0.zip
-rwxr-xr-x 1 katus 197609 28688288 Dec 13 2024 python-3.13.1-amd64.exe*
-rwxr-xr-x 1 katus 197609 26507904 Nov 23 2023 python-3.12.0-amd64.exe*
-rw-r--r-- 1 katus 197609 616451 Jan 27 2024 pxfuel.jpg
-rw-r--r-- 1 katus 197609 32239616 Oct 29 19:45 node-v24.11.0-x64.msi
-rw-r--r-- 1 katus 197609 53358592 Oct 29 19:07 mongosh-2.5.8-x64.msi
-rw-r--r-- 1 katus 197609 51255620 Oct 29 19:06 mongosh-2.5.8-win32-x64.zip
-rw-r--r-- 1 katus 197609 792559616 Oct 29 18:51 mongodb-windows-x86_64-8.2.1-signed.msi
-rw-r--r-- 1 katus 197609 16752316 Nov 23 2023 mingw-w64-x11.0.0.zip
-rwxr-xr-x 1 katus 197609 86528 Nov 23 2023 'mingw-get-setup.exe*'
-rw-r--r-- 1 katus 197609 1678 Feb 1 22:20 kartavya-key-pair.pem
-rwxr-xr-x 1 katus 197609 295320 Jul 9 2024 'dowebsetup.exe*'
-rwxr-xr-x 1 katus 197609 295320 Jul 28 2024 'dowebsetup (1).exe*'
-rwxr-xr-x 1 katus 197609 226334344 Jul 28 2024 'dotnet-sdk-8.0.303-win-x64.exe*'
-rw-r--r-- 1 katus 197609 282 Nov 10 2024 desktop.ini
-rw-r--r-- 1 katus 197609 97320 Dec 20 2024 brandonfowler.exe-runner-1.1.1.vsix
-rw-r--r-- 1 katus 197609 41705 Nov 12 12:08 be_computer-engineering_semester-5_2024_october_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 139091 Nov 12 12:08 be_computer-engineering_semester-5_2024_may_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 88143 Nov 12 12:08 be_computer-engineering_semester-5_2023_november_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 12661 Feb 20 2025 'WhatsApp Image 2025-02-20 at 17.56.00_6b9704e4.jpg'
-rwxr-xr-x 1 katus 197609 94619344 Nov 23 2023 'VSCodesetupSetup-x64-1.84.2.exe*'
-rwxr-xr-x 1 katus 197609 25397512 Jul 28 2024 'VC_redist.x64.exe*'
drwxr-xr-x 1 katus 197609 0 Dec 6 08:28 'Telegram Desktop/'
-rw-r--r-- 1 katus 197609 1175307 Nov 12 12:12 SureLock_seminar.pdf
-rw-r--r-- 1 katus 197609 140151 Jul 22 2024 Self-Declaration-x84-8mm-marathiform.pdf
drwxr-xr-x 1 katus 197609 0 Aug 4 18:57 Orientations/
-rw-r--r-- 1 katus 197609 1359228 Apr 24 2024 LBM_Updated.docx
-rwxr-xr-x 1 katus 197609 167496664 Aug 31 2024 GitHubDesktopSetup-x64.exe*
-rw-r--r-- 1 katus 197609 703657 Nov 13 13:34 G545W94ApplicationForm.pdf
-rw-r--r-- 1 katus 197609 86007808 Dec 22 18:21 EpicInstaller-19.0.0-6e1daf5a6ad8410dad4fc54e9c601d76.msi
-rwxr-xr-x 1 katus 197609 107226856 Mar 8 2025 'DiscordSetup.exe*'
-rwxr-xr-x 1 katus 197609 107226856 Mar 8 2025 'DiscordSetup (1).exe*'
drwxr-xr-x 1 katus 197609 0 Aug 4 18:52 'Day 1 WELCOME + HTML/'
-rw-r--r-- 1 katus 197609 34792458 Jul 28 2024 'DM - Technical.pdf*'
-rw-r--r-- 1 katus 197609 34792458 Aug 19 2024 'DM - Technical (1).pdf*'
-rw-r--r-- 1 katus 197609 801147 Oct 31 15:47 'DBMSL11.pdf'
-rwxr-xr-x 1 katus 197609 1382960 Nov 15 2023 'ChromeSetup.exe*'
-rw-r--r-- 1 katus 197609 7781185 Jul 23 2024 'CG ASSIGNMENT 1.pdf'
drwxr-xr-x 1 katus 197609 0 Dec 3 2023 'B9CEED6F-ASUSPCAssistant_qmba6cd70vzyy/App/'
-rw-r--r-- 1 katus 197609 7942584 Nov 18 22:40 'AnyDesk.exe*'
-rw-r--r-- 1 katus 197609 1512448 Aug 16 2024 722401.msi
drwxr-xr-x 1 katus 197609 0 Jan 2 23:09 ./
drwxr-xr-x 1 katus 197609 0 Feb 1 22:46 ./
katus@Kartavya MINGW64 ~/downloads
$
```

Step 19:-Type command to change the mode

```
katus@kartavya MINGW64 ~/downloads
$ chmod 400 "kartavya-key-pair.pem"

katus@kartavya MINGW64 ~/downloads
$ ls -lar
total 1886349
-rw-r--r-- 1 katus 197609 10041124 Dec 23 15:05 smart-form-filler-demo.mp4
-rw-r--r-- 1 katus 197609 10041124 Dec 23 15:05 'smart-form-filler-demo (1).mp4'
-rw-r--r-- 1 katus 197609 319472 Dec 23 15:07 smart-form-filler-1.0.0.zip
-rwxr-xr-x 1 katus 197609 28688288 Dec 13 2024 python-3.11.1-and64.exe*
-rwxr-xr-x 1 katus 197609 26507904 Nov 23 2023 python-3.12.0-and64.exe*
-rw-r--r-- 1 katus 197609 616451 Jan 27 2024 pxfuel.jpg
-rw-r--r-- 1 katus 197609 32239616 Oct 29 19:45 node-v24.11.0-x64.msi
-rw-r--r-- 1 katus 197609 53358592 Oct 29 19:07 mongosh-2.5.8-x64.msi
-rw-r--r-- 1 katus 197609 51255620 Oct 29 19:06 mongosh-2.5.8-win32-x64.zip
-rw-r--r-- 1 katus 197609 792559616 Oct 29 18:51 mongodb-windows-x86_64-8.2.1-signed.msi
-rw-r--r-- 1 katus 197609 16752516 Nov 23 2023 mingw-w64-v11.0.0.zip
-rwxr-xr-x 1 katus 197609 86528 Nov 23 2023 'mingw_get_setup.exe*'
-r--r--r-- 1 katus 197609 1678 Feb 1 22:20 kartavya-key-pair.pem
-rwxr-xr-x 1 katus 197609 295320 Jul 9 2024 'dowebsetup.exe*'
-rwxr-xr-x 1 katus 197609 295320 Jul 28 2024 'dowebsetup (1).exe*'
-rwxr-xr-x 1 katus 197609 226334344 Jul 28 2024 'dotnet-sdk-8.0.303-win-x64.exe*'
-rw-r--r-- 1 katus 197609 282 Nov 10 2024 desktop.ini
-rw-r--r-- 1 katus 197609 97320 Dec 20 2024 brandonfowler.exe-runner-1.1.1.vsix
-rw-r--r-- 1 katus 197609 41705 Nov 12 12:08 be_computer-engineering_semester-5_2024_october_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 139091 Nov 12 12:08 be_computer-engineering_semester-5_2024_may_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 88143 Nov 12 12:08 be_computer-engineering_semester-5_2023_november_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 1312661 Feb 20 2025 'WhatsApp Image 2025-02-20 at 17.56.00_6b9704e4.jpg'
-rwxr-xr-x 1 katus 197609 94619344 Nov 23 2023 'VSCodiumSetup-x64-1.84.2.exe*'
-rwxr-xr-x 1 katus 197609 25397512 Jul 28 2024 'VC_redist.x64.exe*'
drwxr-xr-x 1 katus 197609 0 Dec 6 08:28 'Telegram Desktop/'
-rw-r--r-- 1 katus 197609 1175507 Nov 12 12:12 SureBlock.seminar.pdf
-rw-r--r-- 1 katus 197609 140151 Jul 22 2024 Self-Declaration-रतन-भरत-marathiform.pdf
-rw-r--r-- 1 katus 197609 0 Aug 4 18:57 Orientations/
-rw-r--r-- 1 katus 197609 1359228 Apr 24 2024 LBM_Updated.docx
-rwxr-xr-x 1 katus 197609 167496664 Aug 31 2024 GitHubDesktopSetup-x64.exe*
-rw-r--r-- 1 katus 197609 703657 Nov 11 13:34 G545W94ApplicationForm.pdf
-rw-r--r-- 1 katus 197609 86007808 Dec 22 18:21 EpicInstaller-19.0.0-6e1daf3a6ad8410dad4fc54e9c601d76.msi
-rwxr-xr-x 1 katus 197609 107226856 Mar 8 2025 'DiscordSetup.exe*'
-rwxr-xr-x 1 katus 197609 107226856 Mar 8 2025 'DiscordSetup (1).exe*'
drwxr-xr-x 1 katus 197609 0 Aug 4 18:52 'Day 1 WELCOME + HTML/'
-rw-r--r-- 1 katus 197609 34792458 Jul 28 2024 'DM - Technical.pdf*'
-rw-r--r-- 1 katus 197609 34792458 Aug 19 2024 'DM - Technical (1).pdf*'
-rw-r--r-- 1 katus 197609 801147 Oct 31 15:47 'DBMSL[1].pdf*'
-rwxr-xr-x 1 katus 197609 1382960 Nov 15 2023 'ChromeSetup.exe*'
-rw-r--r-- 1 katus 197609 7781185 Jul 23 2024 'CG ASSIGNMENT 1.pdf*'
drwxr-xr-x 1 katus 197609 0 Dec 3 2023 'B9CECEDF_ASUSPCAssistant_qmba6cd70vzyy!App/'
-rwxr-xr-x 1 katus 197609 7942584 Nov 18 22:40 'AnyDesk.exe*'
-rw-r--r-- 1 katus 197609 1512448 Aug 16 2024 7z2401.msi
drwxr-xr-x 1 katus 197609 0 Jan 2 23:09 './'
drwxr-xr-x 1 katus 197609 0 Feb 1 22:46 './'
```

Step 20:-Check changed permission

```
katus@kartavya MINGW64 ~/downloads
$ chmod 400 "kartavya-key-pair.pem"

katus@kartavya MINGW64 ~/downloads
$ ls -lar
total 1886349
-rw-r--r-- 1 katus 197609 10041124 Dec 23 15:05 smart-form-filler-demo.mp4
-rw-r--r-- 1 katus 197609 10041124 Dec 23 15:05 'smart-form-filler-demo (1).mp4'
-rw-r--r-- 1 katus 197609 319472 Dec 23 15:07 smart-form-filler-1.0.0.zip
-rwxr-xr-x 1 katus 197609 28688288 Dec 13 2024 python-3.11.1-and64.exe*
-rwxr-xr-x 1 katus 197609 26507904 Nov 23 2023 python-3.12.0-and64.exe*
-rw-r--r-- 1 katus 197609 616451 Jan 27 2024 pxfuel.jpg
-rw-r--r-- 1 katus 197609 32239616 Oct 29 19:45 node-v24.11.0-x64.msi
-rw-r--r-- 1 katus 197609 53358592 Oct 29 19:07 mongosh-2.5.8-x64.msi
-rw-r--r-- 1 katus 197609 51255620 Oct 29 19:06 mongosh-2.5.8-win32-x64.zip
-rw-r--r-- 1 katus 197609 792559616 Oct 29 18:51 mongodb-windows-x86_64-8.2.1-signed.msi
-rw-r--r-- 1 katus 197609 16752516 Nov 23 2023 mingw-w64-v11.0.0.zip
-rwxr-xr-x 1 katus 197609 86528 Nov 23 2023 'mingw_get_setup.exe*'
-r--r--r-- 1 katus 197609 1678 Feb 1 22:20 kartavya-key-pair.pem
-rwxr-xr-x 1 katus 197609 295320 Jul 9 2024 'dowebsetup.exe*'
-rwxr-xr-x 1 katus 197609 295320 Jul 28 2024 'dowebsetup (1).exe*'
-rwxr-xr-x 1 katus 197609 226334344 Jul 28 2024 'dotnet-sdk-8.0.303-win-x64.exe*'
-rw-r--r-- 1 katus 197609 282 Nov 10 2024 desktop.ini
-rw-r--r-- 1 katus 197609 97320 Dec 20 2024 brandonfowler.exe-runner-1.1.1.vsix
-rw-r--r-- 1 katus 197609 41705 Nov 12 12:08 be_computer-engineering_semester-5_2024_october_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 139091 Nov 12 12:08 be_computer-engineering_semester-5_2024_may_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 88143 Nov 12 12:08 be_computer-engineering_semester-5_2023_november_database-management-systems-dms-pattern-2019.pdf
-rw-r--r-- 1 katus 197609 1312661 Feb 20 2025 'WhatsApp Image 2025-02-20 at 17.56.00_6b9704e4.jpg'
-rwxr-xr-x 1 katus 197609 94619344 Nov 23 2023 'VSCodiumSetup-x64-1.84.2.exe*'
-rwxr-xr-x 1 katus 197609 25397512 Jul 28 2024 'VC_redist.x64.exe*'
drwxr-xr-x 1 katus 197609 0 Dec 6 08:28 'Telegram Desktop/'
-rw-r--r-- 1 katus 197609 1175507 Nov 12 12:12 SureBlock.seminar.pdf
-rw-r--r-- 1 katus 197609 140151 Jul 22 2024 Self-Declaration-रतन-भरत-marathiform.pdf
-rw-r--r-- 1 katus 197609 0 Aug 4 18:57 Orientations/
-rw-r--r-- 1 katus 197609 1359228 Apr 24 2024 LBM_Updated.docx
-rwxr-xr-x 1 katus 197609 167496664 Aug 31 2024 GitHubDesktopSetup-x64.exe*
-rw-r--r-- 1 katus 197609 703657 Nov 11 13:34 G545W94ApplicationForm.pdf
-rw-r--r-- 1 katus 197609 86007808 Dec 22 18:21 EpicInstaller-19.0.0-6e1daf3a6ad8410dad4fc54e9c601d76.msi
-rwxr-xr-x 1 katus 197609 107226856 Mar 8 2025 'DiscordSetup.exe*'
-rwxr-xr-x 1 katus 197609 107226856 Mar 8 2025 'DiscordSetup (1).exe*'
drwxr-xr-x 1 katus 197609 0 Aug 4 18:52 'Day 1 WELCOME + HTML/'
-rw-r--r-- 1 katus 197609 34792458 Jul 28 2024 'DM - Technical.pdf*'
-rw-r--r-- 1 katus 197609 34792458 Aug 19 2024 'DM - Technical (1).pdf*'
-rw-r--r-- 1 katus 197609 801147 Oct 31 15:47 'DBMSL[1].pdf*'
-rwxr-xr-x 1 katus 197609 1382960 Nov 15 2023 'ChromeSetup.exe*'
-rw-r--r-- 1 katus 197609 7781185 Jul 23 2024 'CG ASSIGNMENT 1.pdf*'
drwxr-xr-x 1 katus 197609 0 Dec 3 2023 'B9CECEDF_ASUSPCAssistant_qmba6cd70vzyy!App/'
-rwxr-xr-x 1 katus 197609 7942584 Nov 18 22:40 'AnyDesk.exe*'
-rw-r--r-- 1 katus 197609 1512448 Aug 16 2024 7z2401.msi
drwxr-xr-x 1 katus 197609 0 Jan 2 23:09 './'
drwxr-xr-x 1 katus 197609 0 Feb 1 22:46 './'
```

Step 21:- Connect (log in) to an AWS EC2 instance using SSH and check the files present in Instance

```
ec2-user@ip-172-31-34-111:~$ ls -la
total 12
drwxr-xr-x 2 ec2-user ec2-user 29 Feb 16:51 .ssh
-rw-r--r-- 1 ec2-user ec2-user 492 Jan 28 2023 .bashrc
-rw-r--r-- 1 ec2-user ec2-user 141 Jan 28 2023 .bash_profile
-rw-r--r-- 1 ec2-user ec2-user 18 Jan 28 2023 .bash_logout
drwxr-xr-x 3 root root 22 Feb 16:51 ..
drwxr-xr-x 3 ec2-user ec2-user 74 Feb 16:51 .

ec2-user@ip-172-31-34-111:~$ cat /etc/passwd
katus:!:kartavya:197609:25397512:Jul:28:2024:VC_redist.x64.exe:
drwxr-xr-x 1 katus 197609 0 Dec 6 08:28 Telegram Desktop/
-rw-r--r-- 1 katus 197609 1175307 Nov 12 12:12 Sure8lock-seminar.pdf
-rw-r--r-- 1 katus 197609 140151 Jul 22 2024 Self-Declaration-vr8-8mm-marathiform.pdf
drwxr-xr-x 1 katus 197609 0 Aug 4 18:57 Orientations/
-rw-r--r-- 1 katus 197609 1339228 Apr 24 2024 LBN_Updated.docx
-rwxr-xr-x 1 katus 197609 167496664 Aug 31 2024 GithubDesktopSetup-x64.exe
-rw-r--r-- 1 katus 197609 703657 Nov 11 13:34 G545w94ApplicationForm.pdf
-rw-r--r-- 1 katus 197609 86007808 Dec 22 18:21 EpicInstaller-19.0.0-6e1da75a6ad8410dad4fc54e9c601d76.msi
-rwxr-xr-x 1 katus 197609 107226856 Mar 8 2025 DiscordSetup.exe
-rwxr-xr-x 1 katus 197609 107226836 Mar 8 2025 DiscordSetup (1).exe
drwxr-xr-x 1 katus 197609 0 Aug 4 18:52 Day 1 WELCOME + HTML/
-rw-r--r-- 1 katus 197609 34792458 Jul 28 2024 DM - Technical.pdf
-rw-r--r-- 1 katus 197609 34792458 Aug 19 2024 DM - Technical (1).pdf
-rw-r--r-- 1 katus 197609 801147 Oct 31 15:47 DMESL(1).pdf
-rwxr-xr-x 1 katus 197609 1382960 Nov 15 2023 ChromeSetup.exe
-rw-r--r-- 1 katus 197609 7781185 Jul 23 2024 CG ASSIGNMENT 1.pdf
drwxr-xr-x 1 katus 197609 0 Dec 3 2023 R06CEDE8_ASUSPCAssistant_qmba6cd70vzyy/App/
-rwxr-xr-x 1 katus 197609 7942584 Nov 18 22:40 AnyDesk.exe
-rw-r--r-- 1 katus 197609 1512448 Aug 16 2024 722401.msi
drwxr-xr-x 1 katus 197609 0 Jan 2 23:09 ../
drwxr-xr-x 1 katus 197609 0 Feb 1 22:46 ../

katus@kartavya:~/Downloads$ ssh -i "kartavya-key-pair.pem" ec2-user@ec2-16-171-151-85.eu-north-1.compute.amazonaws.com
Warning: Permanently added 'ec2-16-171-151-85.eu-north-1.compute.amazonaws.com' (64:ff9b:10ab:9755) to the list of known hosts.
ED25519 key fingerprint is: SHA256:tana2y98Lsh93ShQwV48rP2hdu10gr8d.dcafu0
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-16-171-151-85.eu-north-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

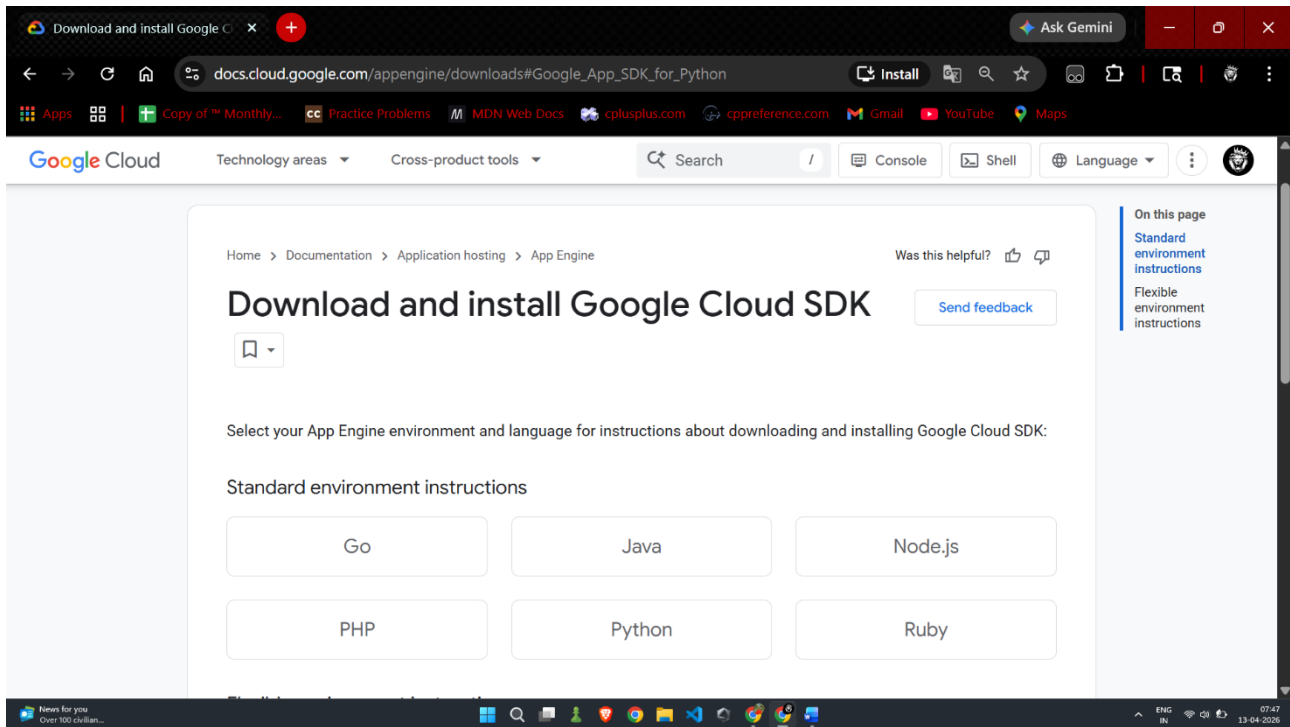
[ec2-user@ip-172-31-34-111 ~]$ ls -la
```

Installation and configure Google App Engine

Step1 : visit :-

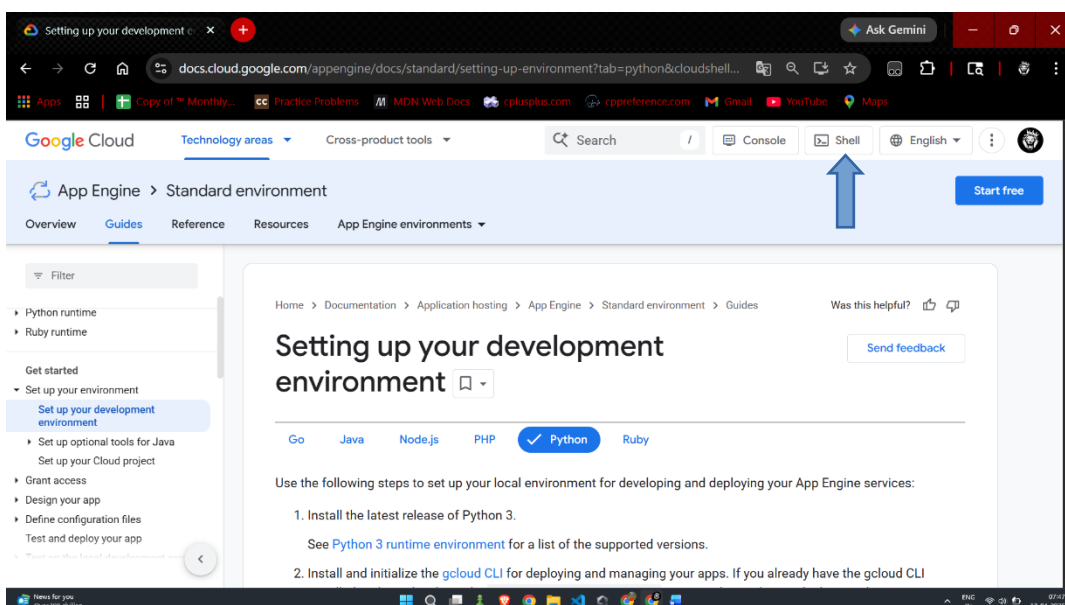
https://docs.cloud.google.com/appengine/downloads#Google_App_SDK_for_Python

Step2 : Select Standard environment instructions as Python



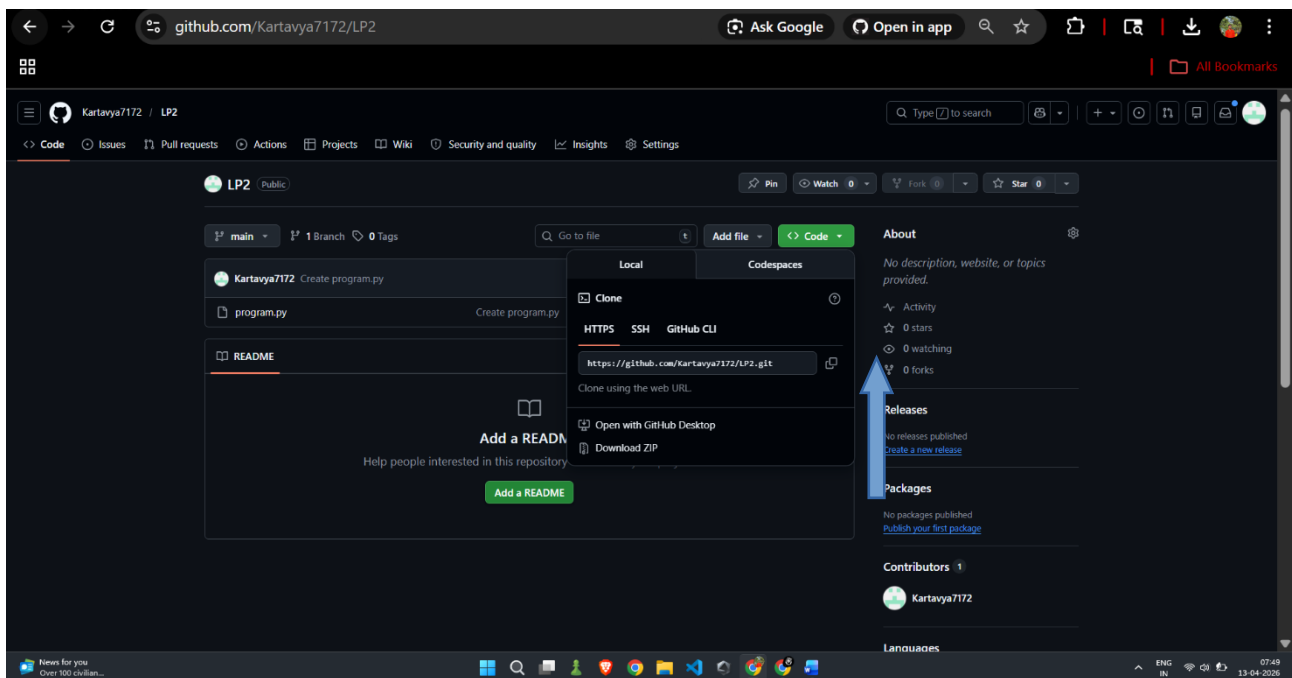
Step3 : SignIn to the google account

Step4 : After Log in go to the shell button present on top right of the screen

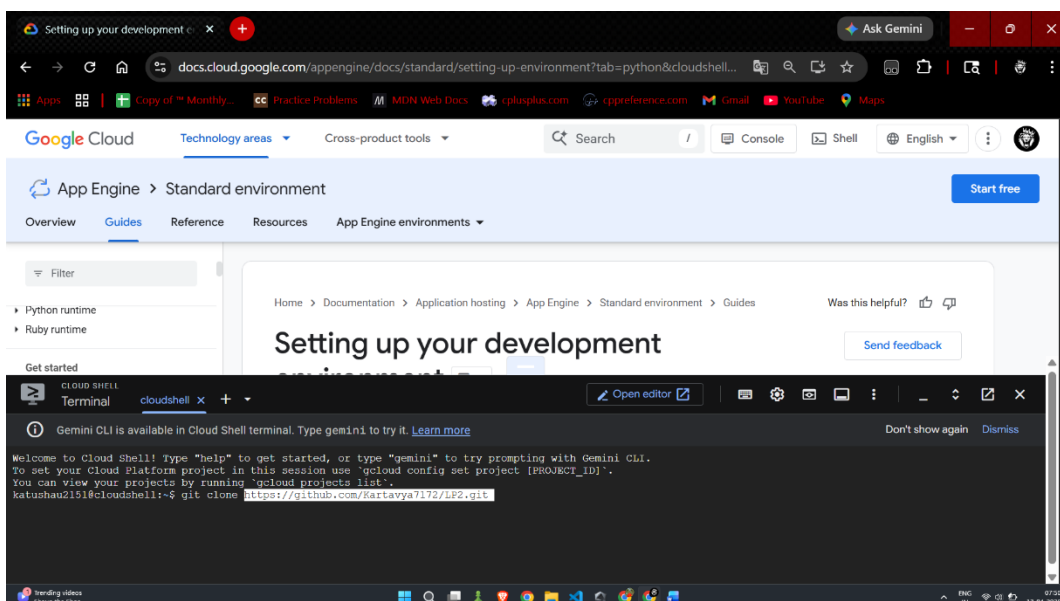


Step5 : You will see the terminal .Go to your github account , create a repository as LP2 (you can give any name). Then create a file in that repository (let say file.py). Write a simple code in that file and commit it.

Step6 : Come back to your repository click on green button and copy the path

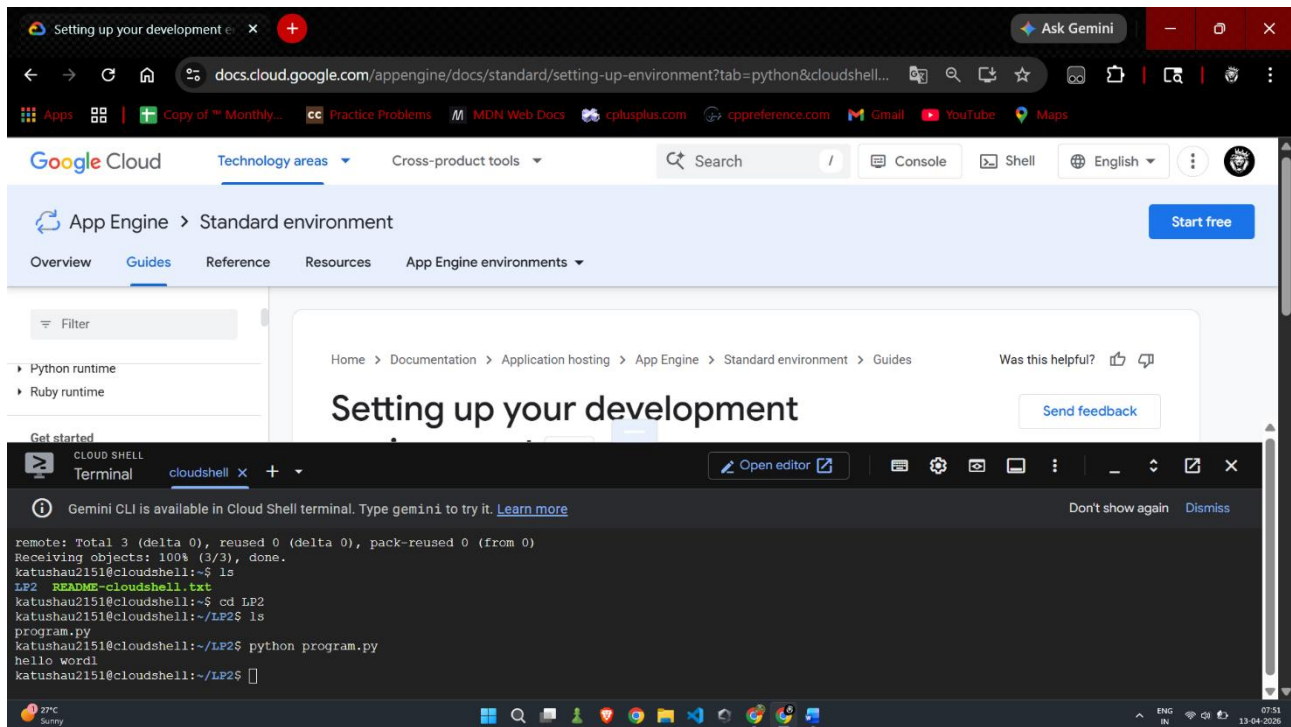


Step7: Come back to terminal and write:- git clone (paste copied path)



Step8 : Now paste these commands one by one

```
ls
cd (your repository name)
ls
python (your file name )
```



Step9 : Done

PRACTICAL-3

Problem statement :

Creating an Application in Salesforce.com using Apex programming Language.

Code:

```
public class LinearSearch {  
    public static void demo(Integer key) {  
        System.debug('Linear Search');  
  
        Boolean found = false;  
        List<Integer> lon = new List<Integer>();  
  
        lon.add(3);  
        lon.add(4);  
        lon.add(5);  
        lon.add(6);  
  
        System.debug('List: ' + lon);  
  
        for (Integer i = 0; i < lon.size(); i++) {  
            if (key == lon[i]) {  
                found = true;  
                break;  
            }  
        }  
  
        if (found) {  
            System.debug('Element Found');  
        } else {  
            System.debug('Element Not Found');  
        }  
    }  
}
```


OUTPUT:

a) When the input number is not found in given numbers:

The screenshot shows the Salesforce Developer Console with the following details:

- Code Editor:** Contains Apex code for a linear search algorithm. Line 10: `lon.add(6);`. Line 12: `System.debug('List: ' + lon);`. Lines 14-19: A `for` loop that iterates through the `lon` list and checks if the current element equals the `key`. If found, it sets `found = true` and breaks the loop. Lines 21-24: An `if (found)` block that calls `System.debug('Element found: ' + key);`. Otherwise, it calls `System.debug('Element not found: ' + key);`.
- Enter Apex Code:** A small window showing the execution of `LinearSearch.demo(777);`.
- Logs:** A table showing the execution log. The log entry is:

User	Application	Operation	Time	Status	Read	Size
Kartavya Torpe	Unknown	/services/data/v66.0/tooling/execute...	4/13/2026, 9:18:56 AM	Success		5.2 KB

The screenshot shows the Salesforce Developer Console with the following details:

- Execution Log:** A table showing the execution log. The log entry is:

Timestamp	Event	Details
09:19:50:005	HEAP_ALLOCATE	[14] Bytes:4
09:19:50:005	VARIABLE_ASSIGNMENT	[14] i3
09:19:50:005	STATEMENT_EXECUTE	[14]
09:19:50:005	HEAP_ALLOCATE	[15] Bytes:8
09:19:50:005	STATEMENT_EXECUTE	[15]
09:19:50:005	HEAP_ALLOCATE	[14] Bytes:4
09:19:50:005	VARIABLE_ASSIGNMENT	[14] i4
09:19:50:005	STATEMENT_EXECUTE	[23]
09:19:50:005	STATEMENT_EXECUTE	[24]
09:19:50:005	HEAP_ALLOCATE	[24] Bytes:17
09:19:50:005	USER_DEBUG	[24] DEBUG:Element Not Found
09:19:50:005	METHOD_EXIT	[1] 01pg5000005WCKj LinearSearch.demo(Integer)
09:19:50:005	SYSTEM_MODE_EXIT	false
09:19:50:005	CUMULATIVE_LIMIT_USAGE	(default)
09:19:50:000	LIMIT_USAGE_EXCEEDED	Number of SOQL queries: 0 out of 100
09:19:50:000	LIMIT_USAGE_EXCEEDED	Number of query rows: 0 out of 50000
09:19:50:000	LIMIT_USAGE_EXCEEDED	Number of SOSL queries: 0 out of 20
09:19:50:000	LIMIT_USAGE_EXCEEDED	Number of DML statements: 0 out of 150
- Logs:** A table showing the execution log. The log entry is:

User	Application	Operation	Time	Status	Read	Size
Kartavya Torpe	Unknown	/services/data/v66.0/tooling/execute...	4/13/2026, 9:19:50 AM	Success		5.27 KB
Kartavya Torpe	Unknown	/services/data/v66.0/tooling/execute...	4/13/2026, 9:18:56 AM	Success		5.2 KB

b) When the input number is found in given numbers:

The screenshot shows the Salesforce Developer Console with the following Apex code:

```
10 lon.add(6);
11
12 System.debug('List: ' + lon);
13
14 for (Integer i = 0; i < lon.size(); i++) {
15     if (key == lon[i]) {
16         found = true;
17         break;
18     }
19 }
20
21 if (found) {
22     System.debug('Element found: ' + key);
23 } else {
24     System.debug('Element not found: ' + key);
25 }
26 }
27 }
```

The 'Enter Apex Code' dialog box contains the text: `LinearSearch.demo(5);`

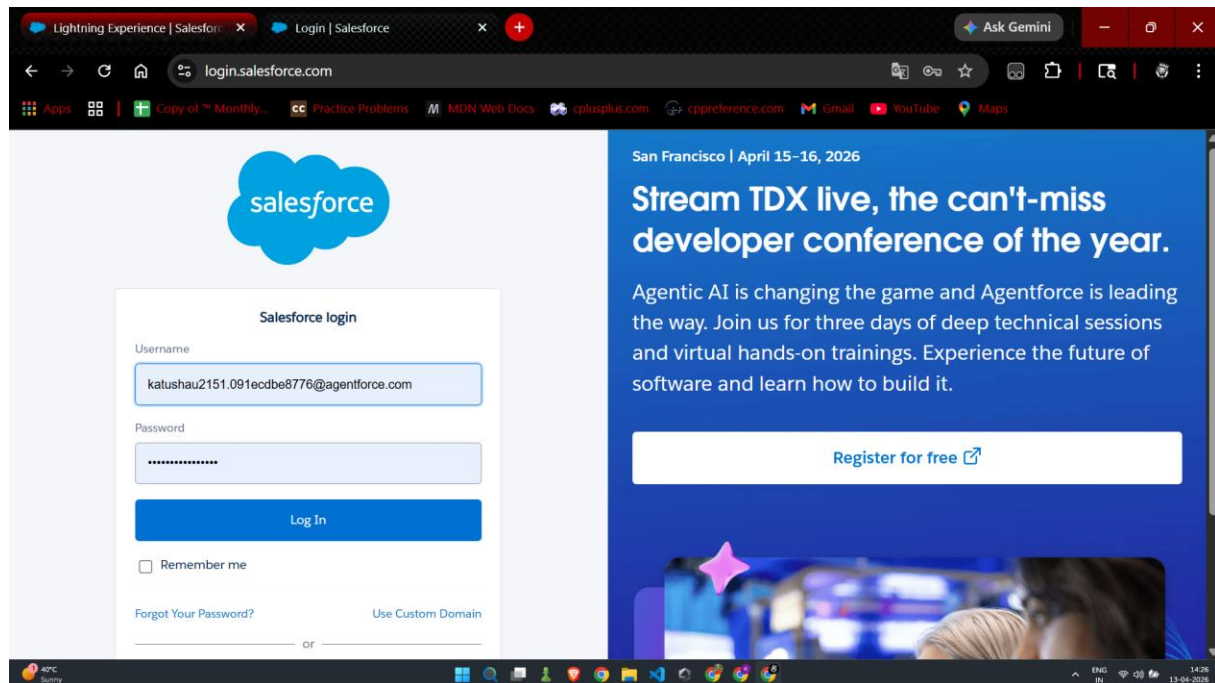
The bottom status bar shows the following buttons: **Execute** and **Execute Highlighted**.

The screenshot shows the Salesforce Developer Console with the following execution log:

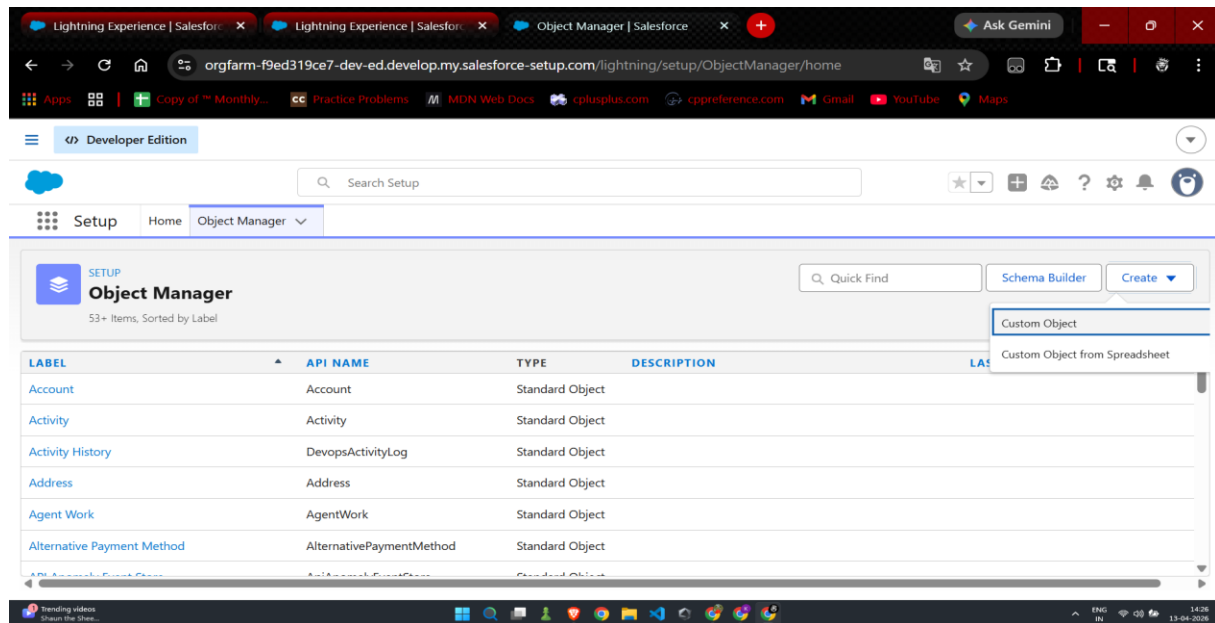
Timestamp	Event	Details
09:18:56:015	STATEMENT_EXECUTE	[16]]
09:18:56:015	HEAP_ALLOCATE	[16]]Bytes:5
09:18:56:015	VARIABLE_ASSIGNMENT	[16]]found=true
09:18:56:015	STATEMENT_EXECUTE	[17]]
09:18:56:015	STATEMENT_EXECUTE	[21]]
09:18:56:015	STATEMENT_EXECUTE	[22]]
09:18:56:015	HEAP_ALLOCATE	[22]]Bytes:13
09:18:56:015	USER_DEBUG	[22]]DEBUG[Element Found
09:18:56:015	METHOD_EXIT	[1]]01pg500005WCqj LinearSearch.demo(Integer)
09:18:56:015	SYSTEM_MODE_CHANGE	false
09:18:56:015	CUMULATIVE_LIMIT_USAGE	(default)]
09:18:56:000	LIMIT_USAGE_REACHED	Number of SOQL queries: 0 out of 100
09:18:56:000	LIMIT_USAGE_REACHED	Number of query rows: 0 out of 50000
09:18:56:000	LIMIT_USAGE_REACHED	Number of SOSL queries: 0 out of 20
09:18:56:000	LIMIT_USAGE_REACHED	Number of DML statements: 0 out of 150
09:18:56:000	LIMIT_USAGE_REACHED	Number of Publish Immediate DML: 0 out of 150
09:18:56:000	LIMIT_USAGE_REACHED	Number of DML rows: 0 out of 10000
09:18:56:000	LIMIT_USAGE_REACHED	Maximum CPU time: 0 out of 10000

The bottom status bar shows the following buttons: **Logs** and **Click here to filter the log**.

Step 1: Log into Salesforce Developer account.



Step 2: Open Salesforce Lightning platform and click on Object Manager => Create => Custom Object.



Step 3: Fill in the required fields and under Optional Features, select **Allow Reports** and **Allow Activities**. Click **Save**.

This screenshot shows the 'New Custom Object' page in Salesforce, specifically the 'Custom Object Definition Edit' section. The page is titled 'New Custom Object' and has a 'SETUP' icon. Below the title, there are 'Save', 'Save & New', and 'Cancel' buttons. The 'Custom Object Information' section contains the following fields:

- Label:** Student_Detail (Example: Account)
- Plural Label:** Student_Details (Example: Accounts)
- Starts with vowel sound:** ☐
- Object Name:** Student_Detail (Example: Account)
- Description:** (Empty text area)

The page also shows a 'Search Setup' bar and a 'Developer Edition' tab. The browser address bar shows the URL: `orgfarm-f9ed319ce7-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/new`.

This screenshot shows the 'New Custom Object' page in Salesforce, specifically the 'Optional Features' and 'Object Classification' sections. The page is titled 'New Custom Object' and has a 'SETUP' icon. Below the title, there are 'Save', 'Save & New', and 'Cancel' buttons. The 'Optional Features' section contains the following fields:

- Data Type:** Text (Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.)
- Optional Features:**
 - ☒ Allow Reports
 - ☒ Allow Activities
 - ☐ Track Field History
 - ☐ Allow in Chatter Groups
 - ☐ Enable Licensing
- Object Classification:**
 - When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)
 - ☒ Allow Sharing
 - ☒ Allow Bulk API Access
 - ☒ Allow Streaming API Access

The page also shows a 'Search Setup' bar and a 'Developer Edition' tab. The browser address bar shows the URL: `orgfarm-f9ed319ce7-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/new`.

Step 4: Now, Click on Fields & Relations => New.

The screenshot shows the Salesforce Developer Edition interface. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. The 'Object Manager' dropdown is open, showing 'Student_Detail'. The main content area is titled 'SETUP > OBJECT MANAGER Student_Detail'. On the left, a sidebar lists various setup options, with 'Fields & Relationships' selected. The main area displays a table of fields for the 'Student_Detail' object, sorted by field label. The table has columns for 'FIELD LABEL', 'FIELD NAME', 'DATA TYPE', 'CONTROLLING FIELD', and 'INDEXED'. The fields listed are 'Created By', 'Last Modified By', 'Owner', and 'Student_Detail Name'. The 'Student_Detail Name' field is highlighted with a blue background.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Last Modified By	LastModifiedBy	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Student_Detail Name	Name	Text(80)		✓

Step 5: Then select option “Email” and click Next-> Next -> Save.

The screenshot shows the Salesforce Developer Edition interface, specifically the 'Fields & Relationships' section for the 'Student_Detail' object. The 'Email' field type is selected in the list on the left. The main area displays the 'Email' field type configuration options, including a description of the field type and a list of related fields. The 'Email' field type is selected, and the 'Next' button is visible at the bottom right.

☐ Master Detail Relationship

☐ External Lookup Relationship

☐ Checkbox

☐ Currency

☐ Date

☐ DateTime

☒ Email

☐ Geolocation

☐ Number

☐ Percent

☐ Phone

☐ Picklist

☐ Restriction (Multi, Select)

Step 6: Similarly. Repeat steps 4 and 5 to add more fields like Phone, Date of Birth. This is how the custom object will have the various fields.

The screenshot shows the Salesforce Setup interface in Developer Edition. The left sidebar lists various setup options, with 'Object Manager' selected. The main content area is titled 'Student_Detail' and shows 'Fields & Relationships' for 7 items, sorted by Field Label. The table below lists the fields:

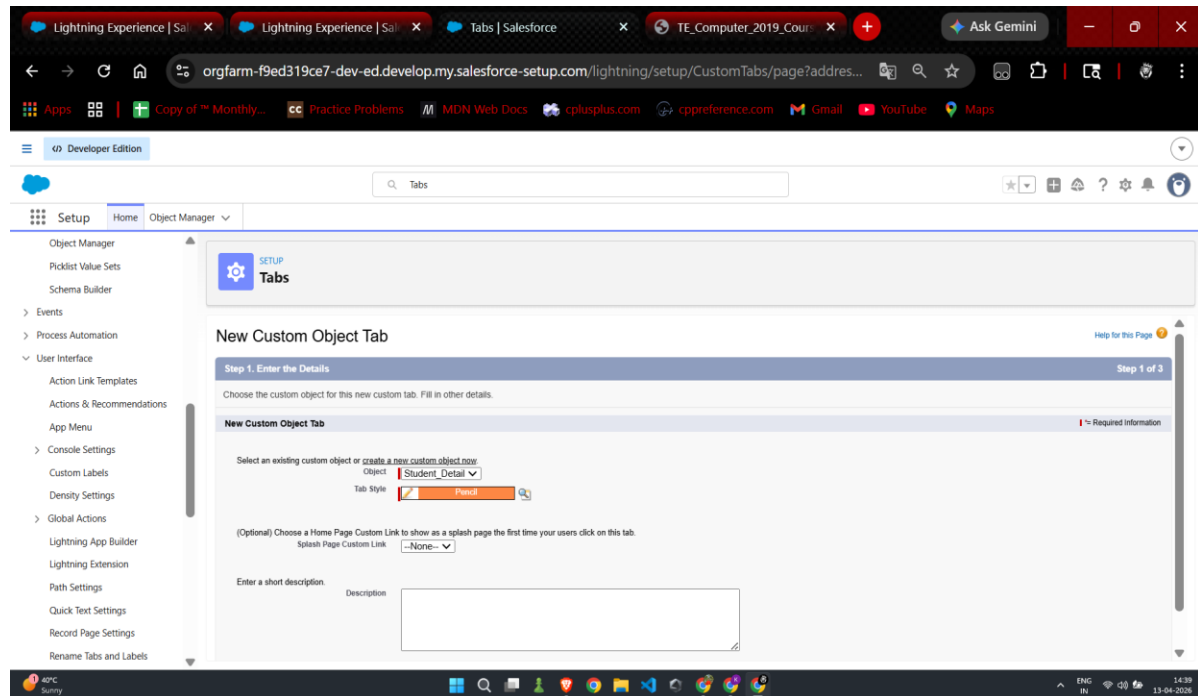
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Student_Detail Name	Name	Text(80)		✓
Student_DOB	Student_DOBs_c	Date/Time		
Student_Email	Student_Emails_c	Email		
Student_Phone	Student_Phones_c	Phone		

Step 7: Now go to Home => search for “Tabs”. Click on New.

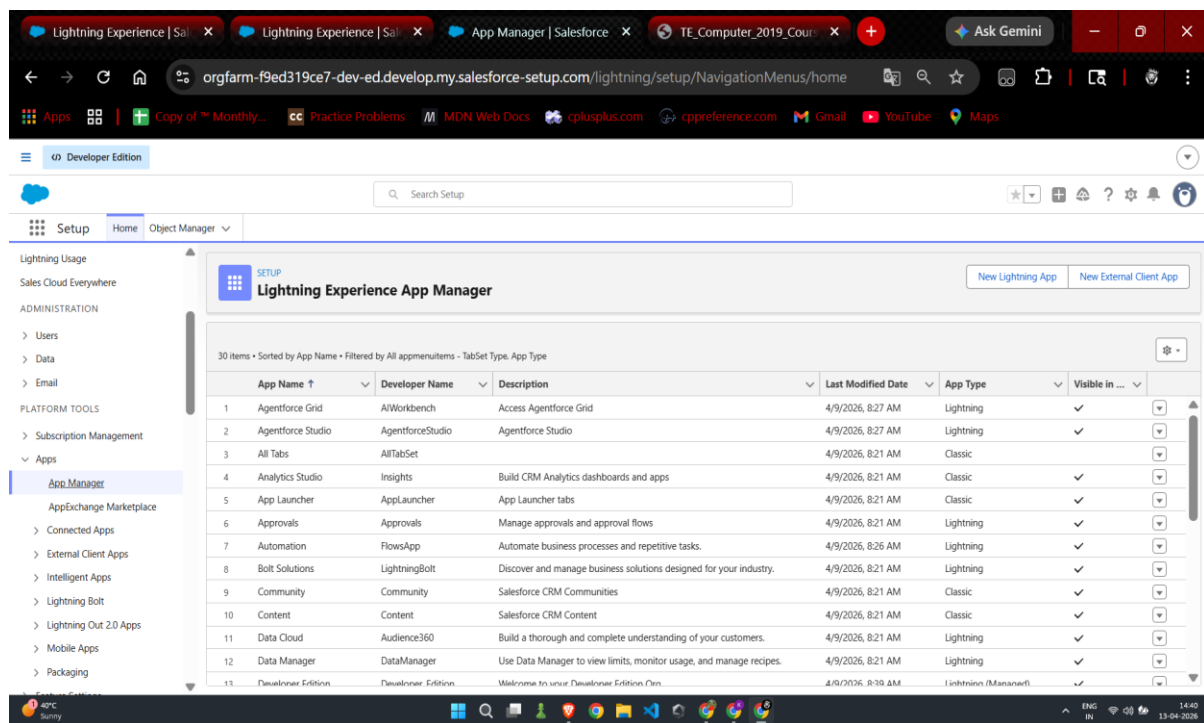
The screenshot shows the Salesforce Setup interface in Developer Edition. The left sidebar lists various setup options, with 'Custom Tabs' selected. The main content area is titled 'Custom Tabs' and shows a list of tabs that have been defined. The table below lists the tabs:

Action	Label	Tab Style	Description
Edit	Get Started with Agentforce	Heart	

Step 8: Enter the Object name and select any icon for tab style. Leave all defaults as it is. Click **Next**, **Next**, and **Save**.



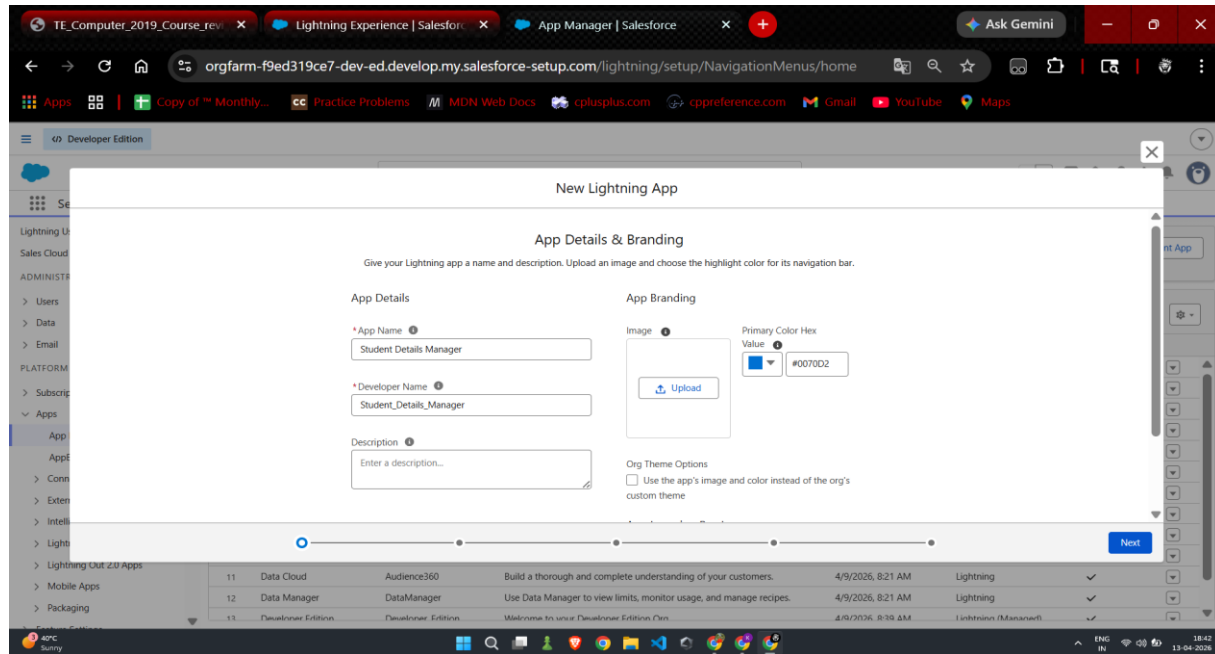
Step 9: In Setup, click **Home**. Enter “App Manager” in Quick Find and select **App Manager**. Click **New Lightning App**.



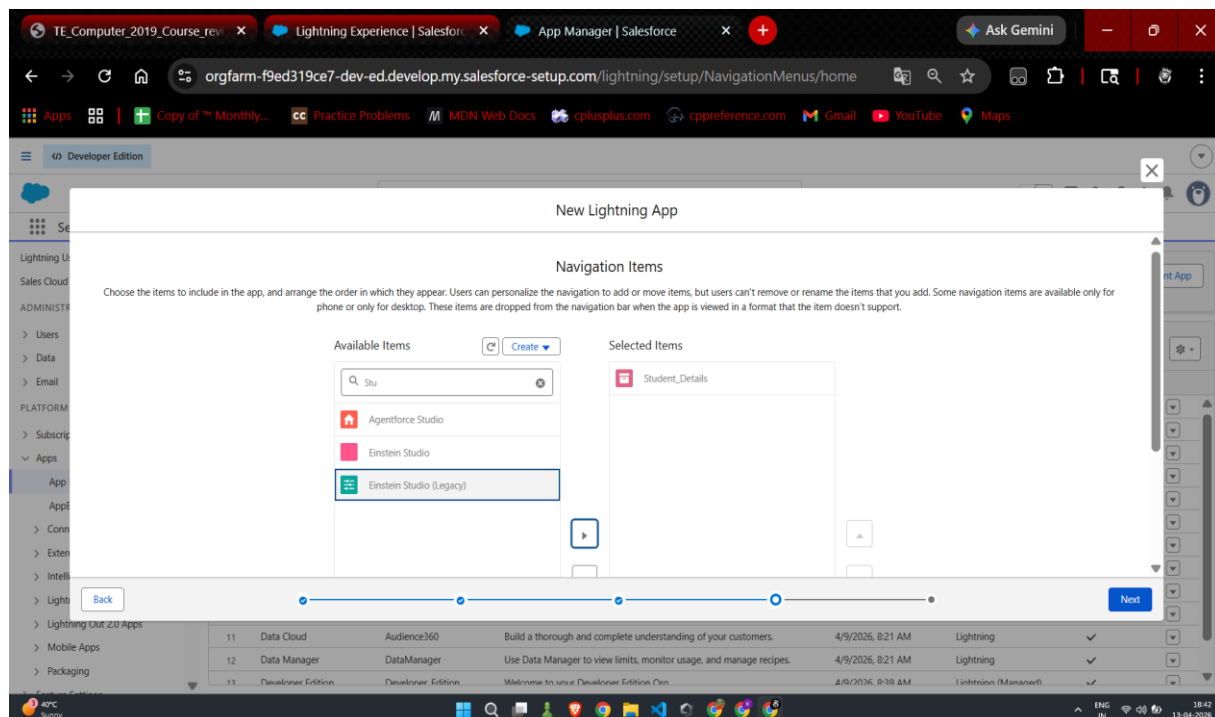
Step 10: Define the new Lightning app as follows:

App Name: Student Details Manager

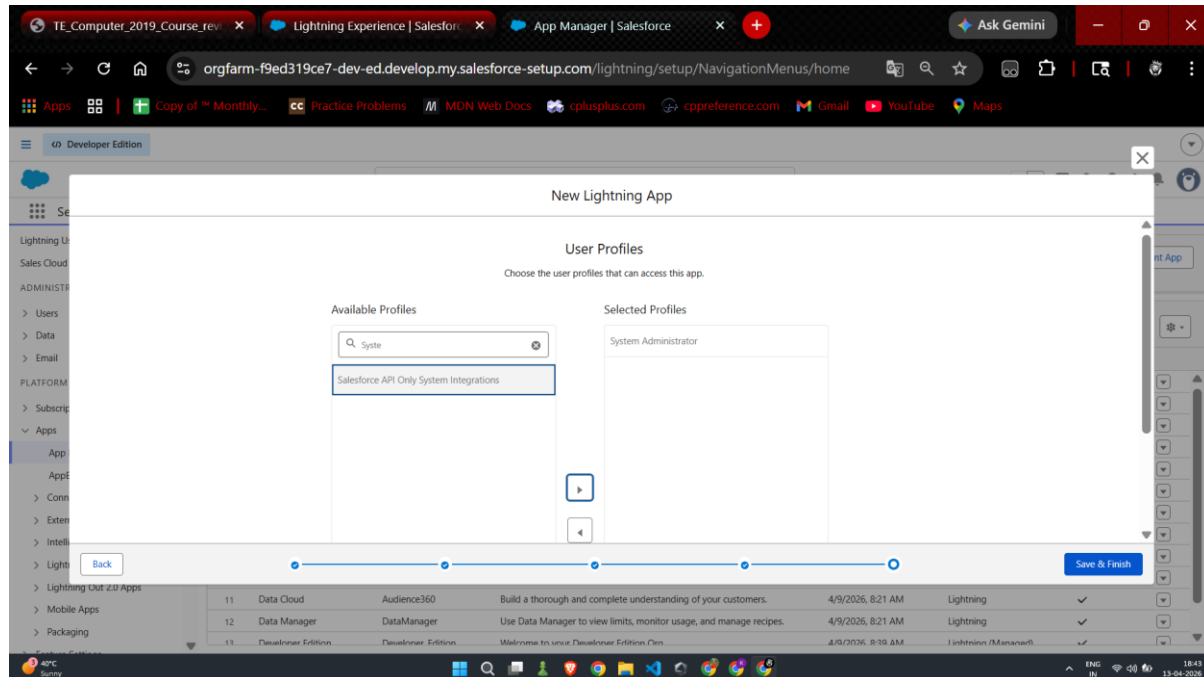
Developer Name: Student_Details_Manager. Click **Next**.



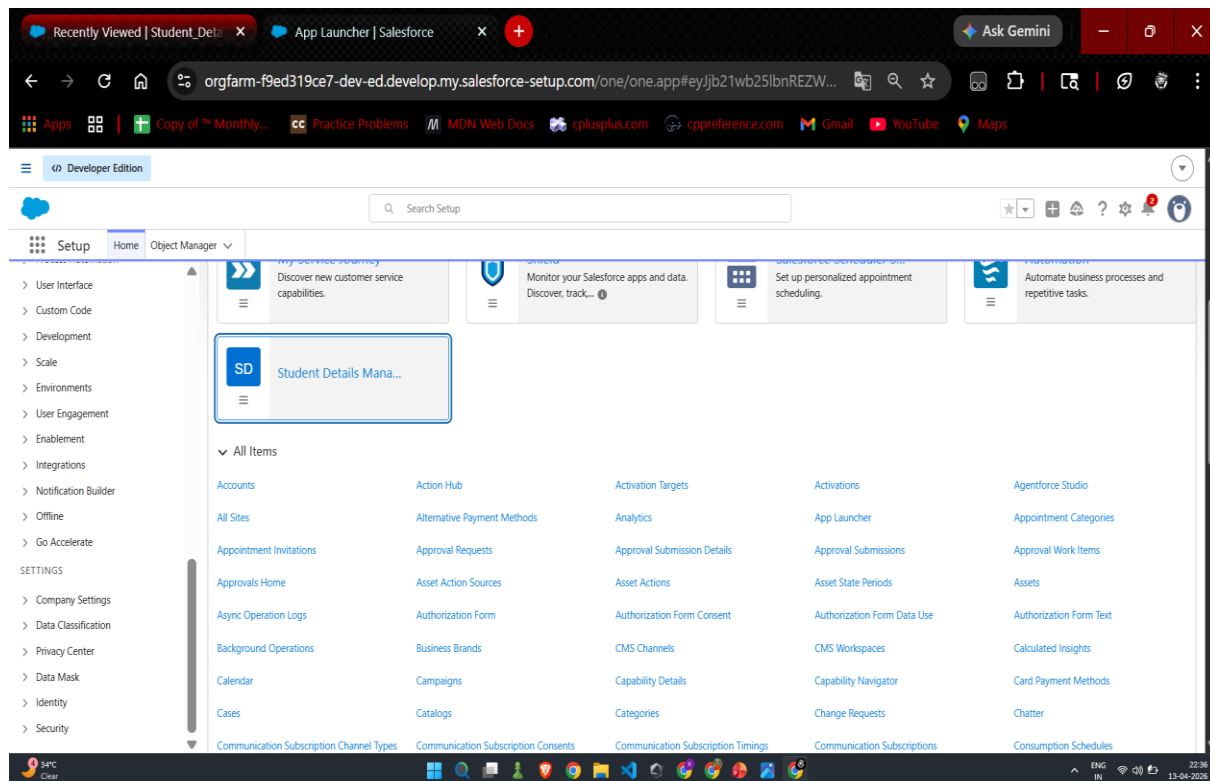
Step 11: On the App Options screen, leave the defaults as is and click **Next**. On the Utility Items screen, leave the defaults as is and click **Next**. On the Navigation Items screen, select **Student_Detail** and move them to the Selected Items box. Then click **Next**.

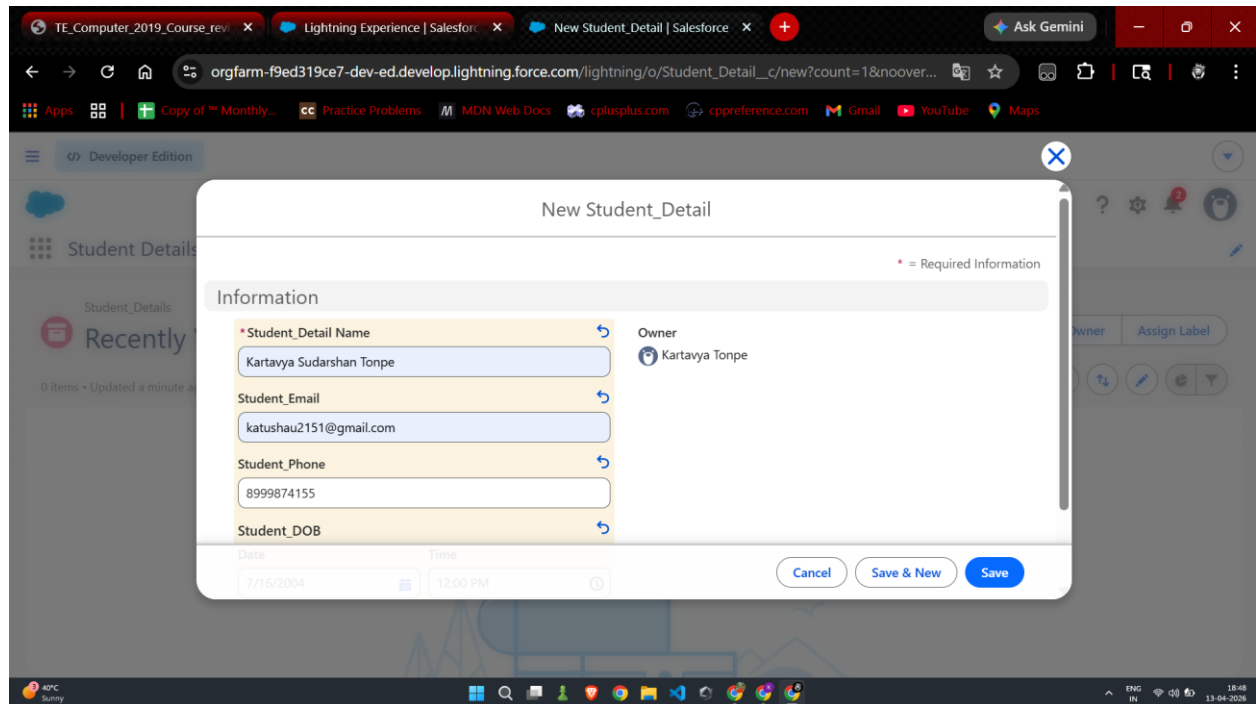


Step 12: On the Assign to User Profiles screen, select **System Administrator** and move it to Selected Profiles. Then click **Save & Finish**.

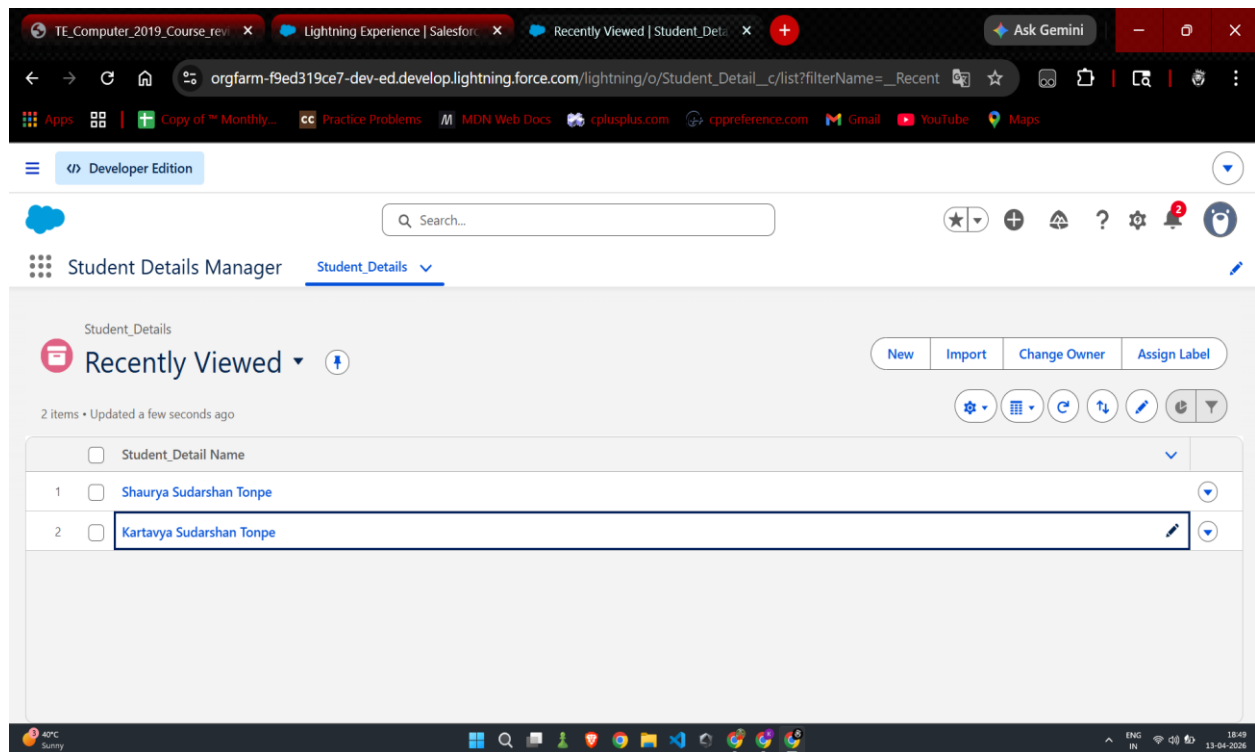


Step 13: Click on App launcher and Open the custom application created.

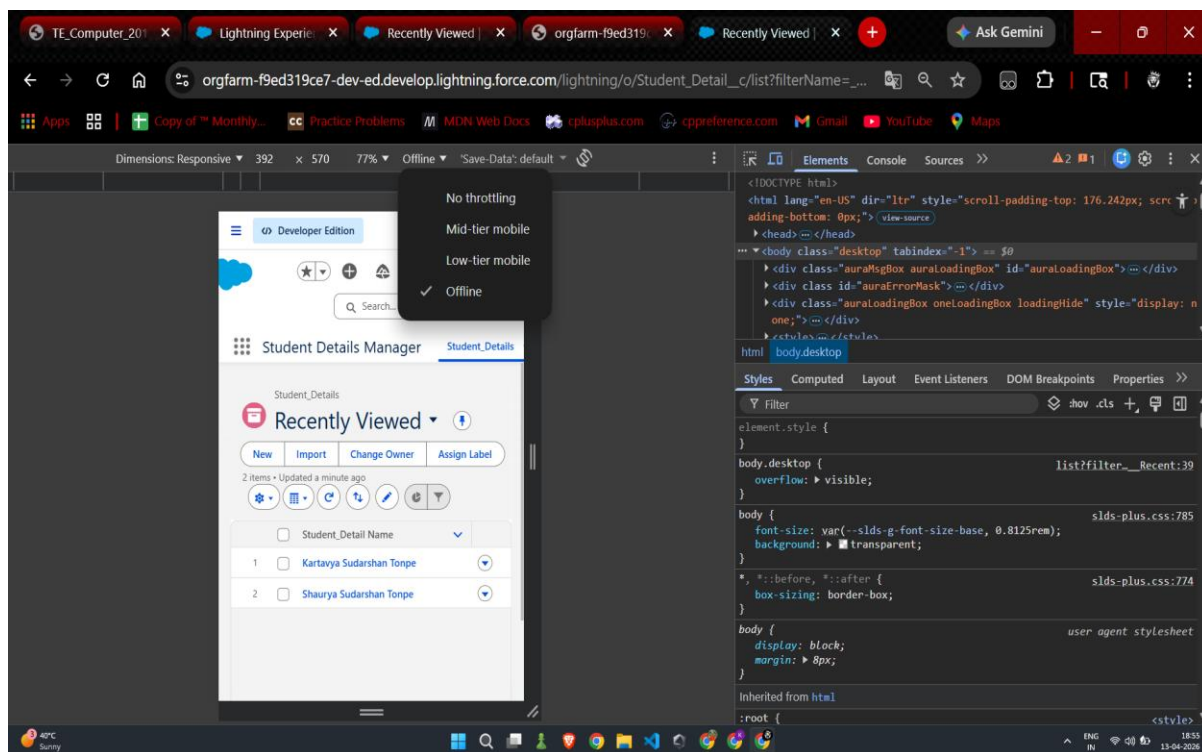
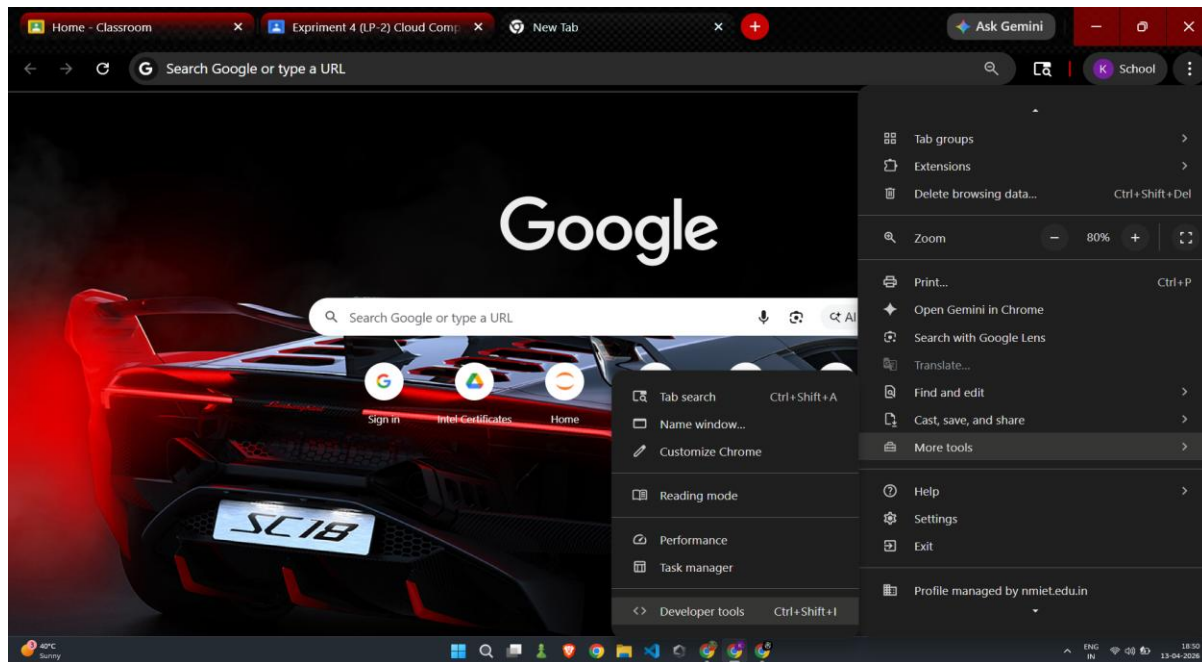


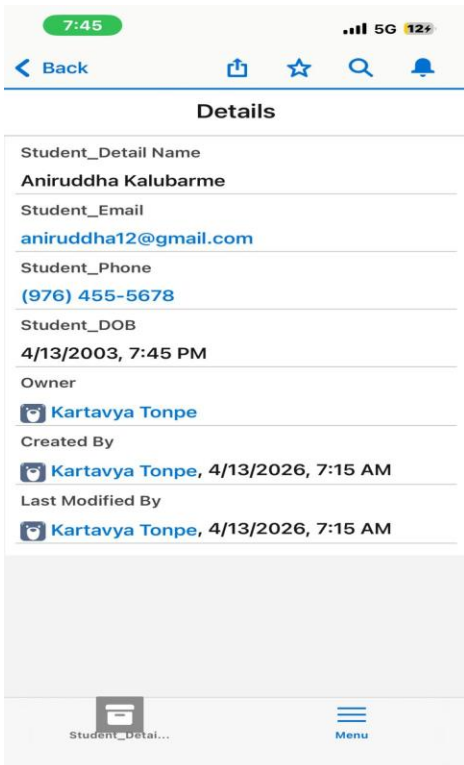
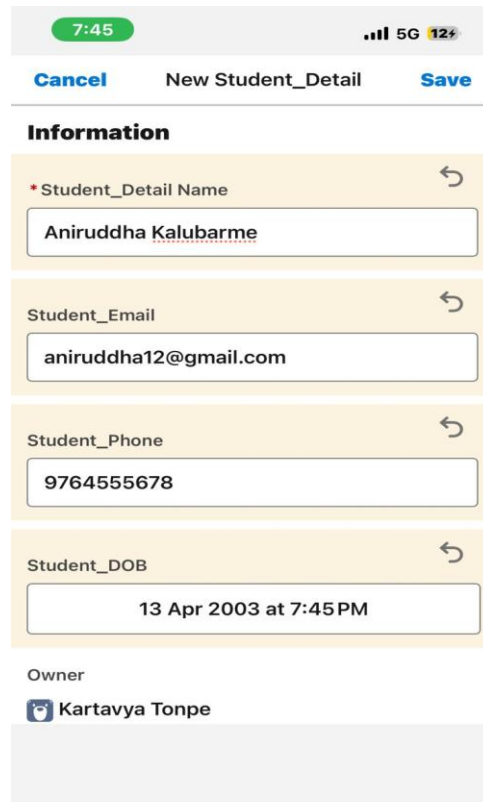
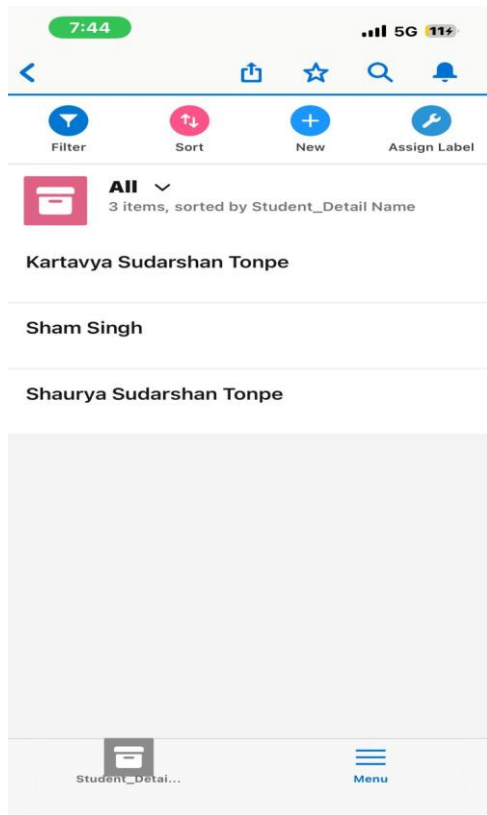


Step 14: Click on Student_Details => New => fill the specified details and copy the URL.



Step 15: Open Google Chrome new Tab => More Tools => Developer Tools and then paste the URL of the application, copied in the previous step.





Conclusion: We have learnt to create custom application using salesforce Lightning platform.